



Extreme events and demand elasticity in energy system models with high renewable penetration and limited foresight



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Abstract

This thesis analyses how commonly used energy modelling assumptions influence the identification of extreme system events and the perceived role of long-duration energy storage (LDES) in highly renewable energy systems. Extreme system events refer to periods of severe system stress, typically characterised by high prices, energy resource scarcity, or load shedding. These indicators are increasingly used to assess energy system robustness. In this study, a stylised energy system model with exclusively variable renewable energy is analysed using both capacity expansion and economic dispatch models. The modelling framework contrasts perfect foresight with limited foresight and inelastic demand with elastic demand, where limited foresight is implemented using a rolling horizon approach and demand flexibility is represented using piece-wise linear demand curves.

The results show that a recently proposed shadow price-based extreme event identification method is capable of identifying events in economic dispatch models that align with periods of high net load, load shedding, and peak utilisation of flexible demand. However, the identified events and the role of LDES differ substantially across modelling assumptions. Under perfect foresight, storage opportunity costs lead to more sustained high prices and a smoothing of scarcity over time, with identified extreme events tending to coincide with high storage utilisation rather than storage depletion. In addition, the identified events in limited foresight models tend to occur earlier in the simulated time period than those identified under perfect foresight. This timing difference suggests that extreme events defined under perfect foresight may lead to different, and potentially misleading conclusions regarding capacity adjustments for a robust system design. Introducing demand elasticity in economic dispatch models further reduces both the frequency and severity of extreme events by absorbing part of the system stress.

The findings demonstrate that extreme event identification and the perceived role of LDES are sensitive to assumptions regarding foresight and demand elasticity. Incorporating limited foresight and elastic demand in energy system models can therefore lead to different conclusions regarding the operation and sizing of long-duration energy storage and support more robust and weather-resilient planning of power systems with high shares of variable renewable energy.

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1 Introduction

1.1 Motivation

Global warming is leading many societies towards a green energy transition. One key challenge in this transition is that, while fossil fuels are relatively easy to store and dispatch on demand, renewable energy generation varies strongly with varying weather conditions. Specifically, wind and solar power depend directly on meteorological conditions, which can result in extended periods of low electricity production during low wind speeds, cloudy conditions, droughts, or prolonged periods of cold weather.

At the same time, electricity demand is becoming increasingly weather-dependent. In an increasingly electrified energy system, electricity is expected to play a central role not only in power supply but also in space heating, industrial processes, and transportation. In particular, cold weather can lead to sharp increases in electricity demand due to electric heating technologies such as heat pumps and direct electric heating. When such periods of low renewable production coincide with high electricity demand over several consecutive days or weeks, the risk of system stress increases significantly.

Such conditions can lead to an imbalance between electricity supply and demand. Given current electricity market designs, this imbalance manifests as very high electricity prices and, in the most severe cases, situations where not all electricity demand can be satisfied. Prolonged or severe occurrences of these conditions can be categorised as extreme system events, characterised by sustained scarcity of energy resources, high system costs, and an increased reliance on dispatchable emergency measures to maintain system balance. During such events, system operators have three main possibilities to attain the system balance:

1. Increase production from dispatchable conventional fossil fuel power plants.
2. Use energy from large storage capacities, such as reservoir hydropower, pumped-storage hydropower, or large underground hydrogen cavern storage¹.
These are generally referred to as long-duration energy storage (LDES).
3. Reduce electricity demand from consumers, in the most severe cases, through load curtailment or load shedding at a very high cost.

The first option is contrary to the objectives of the green transition, which aim to reduce the use and dependency on fossil fuels. The second is expensive to build and maintain enough of, and it requires planning ahead of difficult weather to ensure sufficient energy availability, which is challenging given the uncertain nature of climate and weather dynamics. The third is also very undesirable, since we as societies aim to supply sufficient energy to meet the demand at all times, and it is also expensive to compensate consumers for not being able to meet their demand. The fact that all options involve complications underscores that this is a complex problem with no simple solution.

In the field of energy planning, there is a political priority to expand renewable energy generation. As an example, the European Union has set a target of having a net-zero CO₂ emission in 2050 [1]. This emphasises the importance of considering the risk of extreme system events as societies continue to build more variable renewable energy (VRE). Furthermore, global warming is increasing the frequency and severity of extreme weather [2], suggesting that extreme system events may be

¹Hydrogen can be compressed and stored in large underground caverns formerly used for salt extraction or gas storage.

further exacerbated by future climate changes. Current energy planning is conducted in conjunction with large energy system models that try to identify the best investments to build balanced and resilient energy systems. Thus, energy system models are a key tool for planning how to mitigate the severe consequences of extreme system events.

1.2 Background for energy system modelling

Energy system modelling and optimisation is a field of study where real systems are modelled using physical relations to represent the complex behaviour and dynamics of energy. Furthermore, models can also represent energy markets by linking economic theory with operations research to form mathematical optimisation problems that try to represent both the physical aspect and market dynamics of an energy system. The systems are often so large and complex that it is difficult to represent the entire system and its dynamics in full detail, which necessitates certain assumptions about the behaviour of different elements within the modelling. There are two of these assumptions that are particularly relevant for this thesis (explained in detail in chapter 2). The first is the assumption of perfect foresight. In general terms, this means that the model makes the decisions based on complete knowledge of all data throughout its optimisation period, thus giving the model perfect foresight. This gives the model the possibility of making future decisions based on knowledge and data that would, in reality, otherwise be impossible to have. In weather-dependent systems, this gives the model the advantage of saving energy for periods of scarce weather resources. Weather forecasts are already losing their prediction accuracy when forecasting beyond a 3-day period [3], so to imagine that an energy model can save up energy in an energy storage to use for a bad day 8 months later, is impossible due to the complexity of weather systems. Therefore, operating a system with perfect foresight is an unrealistic representation of reality. However, there are, of course, certain areas, studies, and reasons where the assumption of perfect foresight makes reasonable sense to use, but for the case of reacting to an extreme weather-induced event, the assumption should be questioned.

The second assumption is that energy demand, and specifically electricity demand, is modelled as inelastic with respect to price [4]. This implies that, in both the short and long term, the magnitude of demand does not change even as prices rise. In energy system models, this is typically represented by constraints requiring that electricity demand must always be met (described in chapter 2), even if this leads to extremely high prices. This assumption makes sense in the context that the models aim to optimise new capacity expansion to handle worst-case scenarios. In reality, however, at least part of electricity demand can respond to price signals through demand-side flexibility, such as load shifting, voluntary demand reduction, or automated demand response [5, 6, 7]. During periods of system stress, such flexibility can reduce peak demand and thereby mitigate extreme system events, reducing the need for costly storage dispatch, involuntary load shedding, or costly capacity expansion. As extreme events are often characterised by very high prices, the relationship between demand and price, namely the price elasticity of demand, is particularly relevant to analyse as an alternative to the third option listed above. The direct impact of modelling demand elasticity in combination with extreme event identification is therefore relevant to analyse, as done in this thesis.

1.3 Related literature and research gap

The literature broadly agrees that statistically extreme and rare weather conditions can have severe consequences for weather-dependent energy systems with high penetration of variable renewable energy [8, 9, 10]. The consequences for these systems are, as mentioned, high electricity prices and, in worst cases, the need to shed loads, resulting in some consumers' energy demand not being met. Periods where these things occur to a high degree and accumulate over a longer period (typically beyond a day) can be characterised as extreme system events [11]. Such events might result from compounding non-extreme weather conditions, leading to even greater extremes in such systems [10]. The German term 'dunkelflaute' (dark doldrums) is commonly used to describe a

condition of dark, calm weather leading to low energy production from wind and solar. When this condition occurs during the winter, it is even more severe due to higher electricity demand from increased consumption of electricity-generated heat and generally lower solar energy potential. In energy planning, such events are important to consider, as they drastically increase costs and capacities [12], since the model's response to these periods is either to install new energy capacities or to follow the merit order curve² to a higher energy price. Among the methods to define such events, computing and comparing what is known as the "net load" or "residual load", is often used as a preferred method since it is computationally practical to calculate, and considers both the weather variability, installed capacities and load deviations in the system [13, 14].

Grochowicz, Greevenbroek, and Bloomfield 2024 [9] propose a new identification method where the power system modelling output of shadow prices is used to identify these events. Shadow prices can, in simple terms, be interpreted as the model's electricity price in each node of the system, and are a way of seeing the monetary value of energy in the model at a given point (further explanation in chapter 2). In their study, a capacity expansion model of Europe is solved, and the proposed method is compared with the more traditional net load method, as well as validating events by examining load shedding from running a comparable dispatch model. The findings are that this method actually performs quite well in capturing extreme system events, as it finds the periods that will drive new investments in the system.

In the study, the identified events are validated by comparing with traditional weather data from Euro-Atlantic weather regimes and the NAO³ index. However, the method is tested on a capacity expansion model with perfect foresight and without demand flexibility, which is not a realistic representation of how a real system would respond. Using perfect foresight allows the model to store energy for later usage in times of scarcity, which in reality is only possible to a limited degree.

The general focus on limited foresight in energy planning is often divided into the concepts of long-term planning and operation analysis. For long-term capacity planning, limited foresight can be handled by stochastic models asserting probabilities of realising different outcomes of important uncertain parameters such as the system state and electricity price, as was done recently by Schmidt [15] to assess the importance of long-duration energy storage (LDES). Other studies use the sequential optimisation method of rolling horizon to model limited foresight in the context of transition pathway planning [16]. Nevertheless, the focus of this study remains on modelling limited foresight in the operational context.

While most studies do not focus on limited foresight in relation to extreme system events, Lambert et al. suggest that shorter foresight is better suited to simulating a more realistic system response to unforeseen events or shocks [17].

Furthermore, it is generally acknowledged that demand side management (DSM) or demand flexibility might play a key role in balancing the variability of wind and solar in renewable energy systems [5, 6, 7]. Examples of DSM and demand flexibility are the ability to turn down or switch off loads in a smart way when the system is under stress. While there are different estimates of how flexible demand can be [18], there is a general consensus that electricity demand is somewhat elastic towards prices [4] and not completely inelastic as many models commonly assume by default. In a recent study, Brown, Neumann, and Riepin [19] models flexible demand and limited foresight to analyse price formations in a VRE-based system with no fuel cost. They focus on zero- and scarcity prices as well as how long-duration energy storage contributes to price formation. One of their results is that such a system can still have diverse price dynamics if the demand is modelled with some degree of elasticity.

The identified research gap in the literature is the analysis of extreme event identification in a model without perfect foresight and in a system with flexible demand. Studies have assessed various as-

²Economical term for a curve showing the intersection between price and quantity for both demand and supply of a good. In this case, electricity

³North Atlantic Oscillation

pects of the key concepts underlying extreme event identification, flexible demand, and limited foresight, but a study that links these three remains to be conducted.

Henceforth, in this thesis, I use extreme event identification methods similar to Grochowicz, Greevenbroek, and Bloomfield [9] in a simple stylised energy system model featuring high VRE penetration, and analyse the effects of applying limited foresight and demand flexibility through a modelling approach similar to that of Brown, Neumann, and Riepin [19]. To elaborate further, this thesis seeks to gain a better understanding of how extreme event identification and the supporting role of LDES are influenced by relaxing the common energy modelling assumptions of perfect foresight and inelastic demand. The implications could contribute to a better understanding of how to plan against extreme system events when developing future power systems to be robust and weather-resilient.

1.4 Objective and Research questions

The objective of this thesis is to answer the following research questions:

Main research question

How is extreme system event identification influenced by the assumption of perfect foresight and inelastic demand for highly renewable energy systems, and what are the implications for long-duration energy storage?

Additional research questions

- Are the second extreme event identification method from [9] applicable to dispatch models, and if so, how do the results compare with those from capacity expansion models?
- How does limited foresight affect extreme event identification and the operation of long-duration storage?
- How does demand flexibility affect extreme event identification and the operation of long-duration storage?
- How can changes in event identification under different modelling assumptions be translated into different energy planning conclusions, specifically for the sizing and role of long-duration storage in highly renewable systems?

1.5 Thesis structure and overview

The thesis is structured as follows:

- **Introduction** - Outlines the motivation and background for the study, formulates the main objective and research questions, and clarifies the intended contribution of the thesis.
- **Methodology and optimisation framework** - Presents the theoretical and mathematical foundation of the thesis, including energy system optimization theory, demand representation, extreme event definitions, and the optimisation properties relevant for the analysis.
- **Modelling setup and implementation** - Describes the implemented energy system model, case study selection, techno economic assumptions, scenario design, and the workflow used to conduct the numerical experiments.
- **Results** - Highlights and shows the main results from the capacity expansion and dispatch models, including identified extreme events, storage operation, price formation, and sensitivity analyses.
- **Discussion** - Critically discusses the results, highlights key insights and limitations of the modelling approach, interprets the implications for energy system planning under high re-

newable penetration and outlines recommendations for future research directions to advance this topic.

- **Conclusion** - Summarizes the main findings of the thesis.

2 Methodology and optimisation framework

The methodology of this thesis is based on a quantitative and modelling-based case study approach. A stylised energy system optimisation model is developed and applied to a single country case study in order to systematically compare how extreme system event identification is affected by assumptions of perfect foresight versus limited foresight dispatch, as well as by inelastic and elastic demand representations. The analysis relies on controlled numerical experiments in which these modelling assumptions are varied independently across multiple weather years. Model outputs are evaluated with respect to price formation, long-duration storage operation, and three methods of defining extreme events. The theoretical and mathematical framework underlying the optimisation models and the event identification methods is presented in detail in this chapter.

2.1 Optimisation framework for energy system models

2.1.1 Social welfare maximisation & economic dispatch

Energy system models are all built with a baseline objective of maximising social welfare. The simplest version of such a model is the economic dispatch model for a one-node system. In such a model, social welfare is maximised by maximising the utility from energy demand and minimising the cost of energy production. Typically, the demand assumption is that the utility of demand increases by the same amount for each additional unit of supplied electricity. This is called inelastic demand since it will not change utility no matter the price, as the demand will remain even at very high prices. In many energy models, this is typically assumed as a default [20].

On the supply side, each producer seeks to maximise its own welfare, i.e., their profits. Under the assumption of perfect competition, it can be shown that this yields the same result as minimising the total cost of the entire electricity production. Thus, the results are that in an ideal market with perfect competition, simultaneously maximising demand utility and minimising the cost of supplying energy is the same as maximising social welfare [21]¹.

The simple problem of optimising the energy dispatch of a one-node system is presented below:

$$\max_{g_{i,t}, d_t,} \quad SW = \sum_{t \in \mathcal{T}} U_t(d_t) - \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} c_i^{\text{var}} g_{i,t} \quad (2.1)$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}} g_{i,t} = d_t, \quad \forall t \in \mathcal{T} \quad (2.2)$$

$$0 \leq g_{i,t} \leq G_i, \quad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \quad (2.3)$$

$$d_t \in \mathcal{D}, \quad \forall t \in \mathcal{T} \quad (2.4)$$

$$G, \mathcal{D}, g, c, d \in \mathbb{R}^+ \quad (2.5)$$

Equation 2.1 express maximising social welfare SW through the maximisation of demand utility $U(d_t)$ and cost minimisation of electricity generation $c_i^{\text{var}} g_{i,t}$, where c_i^{var} is the generators marginal production price.

Equation 2.2 is the energy balance equation stating that at any time t , the demand d_t must be met by the sum of generator production $g_{i,t}$ for the total set of generators $\mathcal{I}$.

¹It should be mentioned that this is only true if the used market price is the Lagrange multiplier λ from the energy balance equation. This is explained in section 2.2.2.

Equation 2.3 states how each generators production g_i must stay within its capacity limit G_i in every time step t for the whole optimisation period denoted as the set $\mathcal{T}$.

Equation 2.4 defines the demand d_t as belonging to a feasible set of demand $\mathcal{D}$.

2.1.2 Capacity expansion and dispatch optimisation

In the field of energy system planning, a capacity expansion model is often run together with an economic dispatch optimisation model. The results are that one gets investment cost and installed capacities from the model, which chooses the most feasible technologies based on cost, efficiency, placement, and energy availability of renewable resources. Usually, the capacities are referred to as being exogenously or endogenously determined. Exogenous capacity is fixed, pre-optimisation defined capacities, representing existing energy infrastructure such as power plants, hydroelectric dams, nuclear power, and so forth. Endogenous capacities are those defined solely by the model. For planning models with solely endogenous capacities, the term greenfield optimisation/planning is often used [22, 23]. Greenfield planning refers to building infrastructure and energy assets on a completely empty field. The capacity planning problem run for this thesis is such a model, where no exogenous capacities are defined beforehand.

The optimisation problem for such a capacity expansion problem is similar to the problem described in 2.1.1, but with certain additions. Given the assumption that the demand utility is infinite and inelastic, the optimisation problem's objective function (Equation 2.1) can be reduced to solely focus on minimising cost on the supply site with a hard equality constraint on satisfying the entire demand as a fixed parameter D_t instead of a variable d_t as in the general problem (2.7). This turns the demand utility into a constant term

$$U_t(d_t) = U_t(D_t)$$

Since a constant term cannot be optimised, the only way to maximise social welfare is now to minimise the cost of supplying electricity. This is shown in the following minimisation problem for the capacity expansion problem.

$$\min_{G_i, g_{i,t}} \sum_{t \in \mathcal{T}} c_i^{\text{inv}} G_i + \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} c_i^{\text{var}} g_{i,t} \quad (2.6)$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}} g_{i,t} = D_t, \quad \forall t \in \mathcal{T} \quad (2.7)$$

$$0 \leq g_{i,t} \leq G_i, \quad \forall i, t \in \mathcal{I}, \mathcal{T} \quad (2.8)$$

$$G, g, c, D \in \mathbb{R}^+ \quad (2.9)$$

Equation 2.6 defines the objective function, which minimises the total costs of supplying/generating energy. The first term represents annualised investment costs c_i^{inv} associated with the installed generation capacities G_i . The second term accounts for variable operating costs c_i^{var} incurred by dispatching generation $g_{i,t}$ at time step t . This formulation represents an objective that simultaneously optimises both investment and operational decisions.

Equation 2.7 enforces the energy balance for each time step t , requiring the total generation to meet the electricity demand D_t . Had there been multiple nodes n , there would be such a constraint for each node n . This constraint is what keeps the system in balance at all times. The big difference here compared to the previous balance equation (2.2) is that the demand D_t is a fixed parameter, meaning that the only way this equation can be true is by changing the supply.

Equation 2.8 limits the dispatch $g_{i,t}$ by installed capacity G_i , which is now a variable, thus coupling investment and operational decisions.

Economic dispatch without capacity expansion

This model is also used for the economic dispatch model. The formulation is very similar, the only difference being the exclusion of the capital costs c_i^{inv} and fixing the generator capacity G_i to be a parameter instead of a variable and at the same time removing it from the objective function. An economic dispatch problem is typically how energy markets are solved on a daily basis to form the electricity price and decide which generators will be accepted and dispatched. The mathematical formulation of the problem is shown below.

$$\min_{g_{i,t}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} c_i^{\text{var}} g_{i,t} \quad (2.10)$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}} g_{i,t} = D_t, \quad \forall t \in \mathcal{T} \quad (2.11)$$

$$0 \leq g_{i,t} \leq G_i, \quad \forall i, t \in \mathcal{I}, \mathcal{T} \quad (2.12)$$

$$G, D, g, c \in \mathbb{R}^+ \quad (2.13)$$

Perfect and limited foresight in energy system models

The assumption of perfect foresight explains the model's knowledge during the decision points, specifically what data the model can see and include in the optimisation. The time-varying data are usually weather, demand, price, or system-state data (whether it is positive or negative). In models that assume perfect foresight, all the decisions are taken in one step, and all relevant data are known to the model at that decision point.

This enables the model to plan in the absolutely most cost-efficient way. A central example of this is storing energy at the beginning of the year to utilise it later in the year when the energy is needed. This is the key rationale for challenging this assumption by implementing a limited foresight model.

The modelling of limited foresight can be divided into two (rather simplified) areas: the aforementioned stochastic modelling and sequential optimisation modelling [24]. The key feature of both is that decisions are made at different stages or time steps, with limited data availability, as the data is released to the model step by step.

Limited foresight optimisation using the rolling horizon method

In this thesis, the limited foresight models are run using the rolling horizon method. Intuitively, the rolling horizon approach can be understood as solving a sequence of dispatch optimisation problems over consecutive time windows with limited look-ahead. At each iteration, the model has perfect foresight within a finite horizon of length H , but no information beyond this window. Temporal consistency across the full simulation period is ensured by carrying forward intertemporal state variables between windows.

In Figure 2.1, an example of this is shown for a yearly optimisation with a horizon length of one day. The figure shows how the rolling horizon method performs multiple small optimisations, transferring the optimised variable and state between windows. Whereas a perfect foresight model optimises everything at once.

2.1.3 Cost recovery and long run equilibrium

The optimisation framework used in this thesis is commonly interpreted as representing a long-run equilibrium of a competitive electricity market. Under this interpretation, investment and dispatch

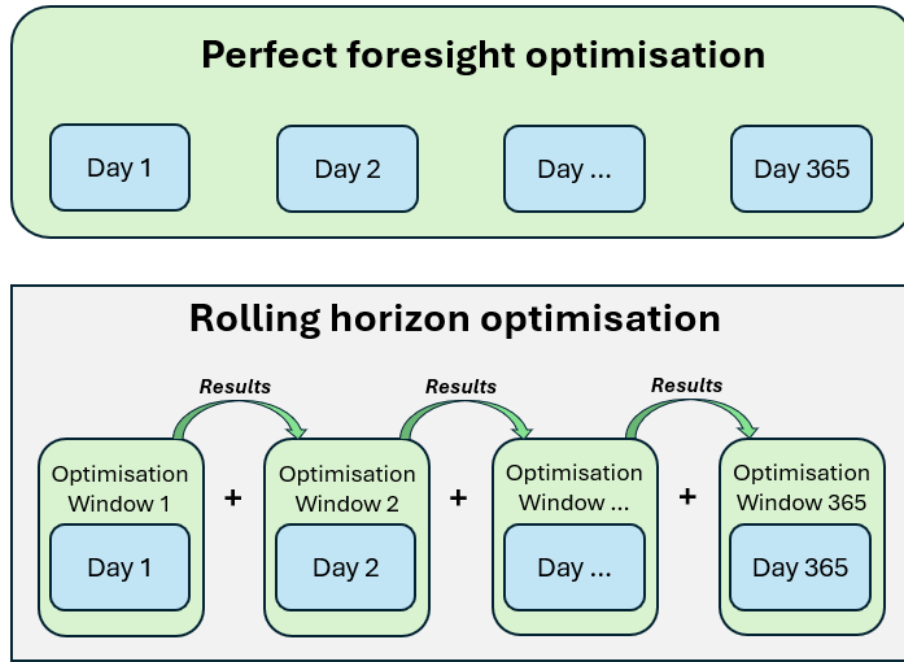


Figure 2.1: Rolling horizon optimisation compared to perfect foresight optimisation. Shown for a yearly optimisation with a horizon of 1 day

decisions emerge endogenously from the optimisation problem and reflect economically rational behaviour under idealised market conditions. In order to motivate the mathematical properties discussed in the following section, the key economic assumptions underlying this interpretation are briefly outlined below.

Perfect competition

The models considered in this thesis assume perfect competition among market participants. This implies that all agents are price takers, possess complete information within the optimisation horizon, and act to maximise their individual economic objectives. No single agent is able to influence market prices, and all technologies are represented by their marginal and fixed costs without strategic behaviour or market power. Under these assumptions, market outcomes can be represented by a centralised optimisation problem that maximises social welfare or, equivalently, minimises total system costs, as we have seen in the previous sections.

Long run equilibrium

The joint optimisation of investment and dispatch decisions corresponds to a long-run equilibrium in which capacity expansion has adjusted to expected operating conditions. In this equilibrium, technologies with lower costs are utilised more frequently, while higher cost technologies are only built and dispatched when they are required to meet demand or system constraints. This implies that only technologies that contribute positively to system performance are invested in and installed. The equilibrium interpretation relies on the mathematical structure of the optimisation problem, which ensures that optimal solutions can be interpreted consistently in economic terms. The additional conditions that enable this interpretation are discussed in the following sections.

Cost recovery

Under long-run equilibrium and perfect competition, a central property of the optimisation outcome is cost recovery. Technologies that are built and operated in the optimal solution are able to recover both their variable operating costs and their fixed investment costs over the optimisation horizon. In the optimisation framework, this condition is not imposed explicitly but arises endogenously over time through the interaction between investment variables, operational constraints, and demand.

2.2 Mathematical properties and solution theory for optimisation problems

2.2.1 Linear optimisation and convexity

Energy system optimisation problems are often formulated as linear programs, as linearity reduces computational complexity and enables the use of efficient and well-established solution algorithms. For linear and convex optimisation problems, solvers are guaranteed to find a global optimum provided that the problem is feasible and bounded.

There exist several classes of algorithms for solving optimisation problems, depending on the mathematical structure of the problem. Linear programs are typically solved using simplex or interior-point methods, which exploit the polyhedral structure of the feasible space of the problem [25].

A key mathematical property underlying the interpretation of linear optimisation models is convexity. In convex optimisation problems, any local optimum is also a global optimum. This property ensures that the solution found by the solver represents a globally optimal system configuration rather than a locally optimal one [26].

Linearity is a sufficient, but not necessary, condition for convexity. As a result, linear optimisation problems are always convex, but convex problems may also be non-linear. In this thesis, the convexity of the optimisation problems ensures that differences in model results can be attributed to differences in modelling assumptions, such as foresight and demand flexibility, rather than to numerical artefacts or convergence to different local optima. This is of great importance for interpreting and comparing the results and for comparing the different models [26].

2.2.2 Dual variables and KKT conditions

Optimisation problems formulated for energy system models can be analysed not only through their primal decision variables, such as generation, storage operation, and installed capacity, but also through their associated dual variables. Following the theory of mathematical optimisation and duality, every problem has a corresponding dual problem [26]. For linear and convex optimisation problems this implies, that every constraint is associated with a corresponding dual variable. This variable measures the marginal change to the objective function resulting from a relaxation of that constraint. Since the objective function represents the economy of the problem, this means that the interpretation of dual variables can help understand the underlying economic properties of our models.

In linear optimisation, the concept of strong duality holds, meaning that the optimal objective value of the primal (main) problem equals that of its dual problem [26]. This property ensures that marginal values derived from the constraints are internally consistent between the two problems and that the dual variables can be interpreted as equilibrium prices under the assumptions of perfect competition and convex costs.

In the context of energy system models, the primal problem represents the physical investment and dispatch decisions, as established in Sections 2.1.1–2.1.2, while the dual variables represent the marginal economic values of scarce resources such as fuels, energy, and generation capacity, commonly referred to as Lagrangian multipliers.

A central example is the dual variable associated with the energy balance constraint (2.7), which has the dual variable λ :

$$\sum_i g_{i,t} + s_t^{\text{dis}} - s_t^{\text{ch}} = D_t \Leftrightarrow \lambda_t \quad \forall t \in \mathcal{T} \quad (2.14)$$

This dual variable represents the marginal cost of supplying an additional unit of electricity at a given time step and is commonly interpreted as the model's shadow price. When solving an economic dispatch problem for an electricity market, these values can be interpreted as the electricity

price. Although the models in this thesis do not represent a real market, the shadow prices from the optimisation results might still be referred to as the model's electricity price in the reporting.

Similarly, dual variables associated with generation capacity constraints represent scarcity rents, reflecting the value of relaxing the capacity limitations during periods of high system stress. These are denoted as μ_{upper} or μ_{lower} .

The validity of these economic interpretations and why we can be sure to use them is ensured by the Karush–Kuhn–Tucker (KKT) conditions, which characterise optimal solutions to convex optimisation problems. For linear and convex problems, the KKT conditions are both necessary and sufficient for optimality [25, 26]. As a result, they provide a link between the mathematical solution of the optimisation problem and its economic interpretation.

The KKT conditions consist of primal feasibility, dual feasibility, and complementary slackness. Primal feasibility ensures that the solution satisfies all physical and operational constraints of the system, while dual feasibility guarantees that the associated dual variables are economically meaningful, in the sense that relaxing a constraint cannot reduce total system cost or welfare.

Complementary slackness links the primal and dual problems by stating that a constraint can only have a non-zero dual value if it is binding at the optimum. As a result, prices and scarcity rents only arise when a physical or operational constraint actively limits the system.[25, 26]

Complementary slackness is particularly important for the analysis of extreme events, as it directly explains the relationship between system stress and price formation. The shadow price λ_t increases when the system is constrained by limited generation, storage availability, or other flexibility options, while prices remain low when sufficient energy is available to the model. During extreme system events, several constraints may become binding simultaneously, resulting in sharp price spikes and high cumulative shadow prices.

In this thesis, dual variables and KKT conditions provide the theoretical foundation for analysing price formation and resource scarcity in relation to extreme system events, as the cumulative behaviour of the shadow prices over the optimisation period is used as a key indicator to identify extreme events, as will be described further in the following section.

2.3 Extreme system event identification

Generally, an extreme system event is when there are prolonged difficulties delivering the demanded energy in a given time period, leading to load shedding and/or high consumer costs [11]. There are three main conditions under which this can happen: The first being if the demand is much higher than normal, reaching the limits of the system's power generation capacity. The second is if the generation is limited due to faults (generator outages, cable disconnections, etc.) or a lack of energy resources. The last and most severe one is when both conditions are present. Due to the increasing presence of varying renewable energy, such as solar PV (photovoltaic) and wind turbines, in our electricity system, periods with low available energy come naturally following the ever-varying weather conditions.

A natural measure of an extreme period commonly used is the concept of net load (also commonly referred to as residual load). This is a measure of the mismatch between VRE generation potential and load. It is defined as:

$$L_t^{\text{net}} = D_t - \sum_{r \in \mathcal{R}} g_{r,t}^{\text{VRE}} \quad \forall t \in \mathcal{T} \quad (2.15)$$

In Equation (2.15), L_t^{net} denotes the net load at time step t , defined as the residual electricity demand that must be supplied by dispatchable generation, storage, or demand flexibility. D_t represents total electricity demand, while $g_{r,t}^{\text{VRE}}$ is the electricity generation from variable renewable energy

technology r at time t . The set $\mathcal{R}$ contains all variable renewable technologies considered, such as wind and solar, and $\mathcal{T}$ denotes the set of time steps in the optimisation horizon.

The term residual load describes the portion of demand that must be supplied by dispatchable generators. Typically, this would be conventional generators such as gas, coal or biomass power plants, but could also be nuclear, hydropower or other types of energy storage. Nevertheless, it will from this point be referred to as the net load.

Another way to look at extreme periods is to identify periods with blackouts or load curtailment. The last resort for the system operator to handle an extreme event is to shut down some of the load, which means that this is also a good indicator of the system being under stress.

In this thesis, such an event type is defined as the period with the highest amount of load shedding across the entire year. This concrete method of how this is defined is described in subsection 3.5.3.

2.3.1 Event identification based on optimisation outputs

One of the main objectives of this thesis is to analyse how the event identification method proposed by Grochowicz, Greevenbroek, and Bloomfield [9] works for dispatch models. The method looks at the output from the optimisation model, more specifically, the shadow prices. The mathematical formulation of the method is:

$$\sum_{n \in N} \sum_{t=t_0}^{t_0+T-1} D_{n,t} \cdot \lambda_{n,t} \geq C \quad (2.16)$$

Where $\lambda_{n,t}$ is the shadow price at time t in node n from the total node set N ; $D_{n,t}$ is the demand at time step and node. C is the cost threshold, above which the accumulated cost must be greater than. Lastly, T and t_0 are the maximum and minimum duration time for an event to occur in. The specific threshold values are presented in subsection 3.5.1.

Explained intuitively, the method states that if a relatively short period has an accumulated cost higher than a given threshold, this will be categorised as an extreme and system-defining event.

2.4 Demand representation and flexibility theory

The general assumption that demand is perfectly inelastic does not hold in reality [27]. The demand utility does have a limit, and the system typically also has a limit. In most systems, this is reflected by having a load shedding cost. This means that the utility of demand will increase until a high price point where the transmission system operator (TSO) will be shedding / curtailing load of the system. This results in the utility function:

$$U(d) = P^{ls} \cdot d \quad (2.17)$$

This means that the utility increases until P^{ls} at which the demand is 100% elastic. This is, however, also a simplification of reality. Some demand will be price sensitive and thus flexible. Instead of having a simple linear utility function for demand like in eq. (2.17), we could also have a more complex function, such as a quadratic or higher polynomial function. This thesis models electricity demand using a quadratic utility function, which yields a linear demand–price relationship.

The following theory follows that presented by Brown et. al. Appendix 1 [19], however, some additional mathematical deductions are included here to help explain the implementation of the specific modelling in this thesis.

2.4.1 Modelling linear and piece-wise linear demand flexibility as load shedding generators

The base of this flexibility modelling is that instead of directly changing the demand utility function in the objective function of the model, the flexible demand will be represented as load shedding generators with a cost relating to the decreasing utility at higher system prices. This is done due to faster convergence and better numerical stability when running the model [19].

The first step is to formulate a quadratic utility function:

$$U(d) = a_d d - \frac{1}{2} b_d d^2 \quad (2.18)$$

Taking the derivative of the utility function gives us the corresponding inverse linear demand curve:

$$U'(d) \Rightarrow p(d) = a_d - b_d d \quad (2.19)$$

To prove that this can also be modelled as a load shedding generator, we must substitute the demand variable d with the baseline demand (mean or peak demand) D minus the generator's dispatch/production variable g :

$$d = D - g \quad (2.20)$$

Substituting this into the utility function gives:

$$U(D - g) = a_d(D - g) - \frac{1}{2} b_d(D - g)^2 \quad (2.21)$$

Expanding gives:

$$U(D - g) = a_d D - \frac{b_d D^2}{2} + (b_d D - a_d)g - \frac{1}{2} b_d g^2 \quad (2.22)$$

To be able to express this as a generator cost, the signs must be flipped, following the objective function (eq 2.1) convention. The constant term $a_d D - \frac{b_d D^2}{2}$ can be excluded as it is not influenced by any variables. This now gives us:

$$C(g) = a_{\text{gen}} \cdot g + b_{\text{gen}} \cdot g^2 \quad (2.23)$$

where

$$a_{\text{gen}} = (a_d - b_d D) \quad \text{and} \quad b_{\text{gen}} = \frac{1}{2} b_d \quad (2.24)$$

With a_{gen} representing the marginal cost and b_{gen} the marginal quadratic cost of the load shedding generator.

What determines the flexibility is how the parameters a_d and b_d are chosen. A good measure of flexibility is price elasticity, which is an economic term used to describe the rate of change in quantity when the price changes by one unit. For a linear demand curve, the elasticity changes along the curve. The price elasticity at a given point can be defined as:

$$\varepsilon_{d,p} = \frac{\partial d}{\partial p} \frac{p_{\text{ref}}}{d_{\text{ref}}} \quad (2.25)$$

where

$$\frac{\partial d}{\partial p} = \frac{1}{p'(d)} = -\frac{1}{b_d} \quad (2.26)$$

such that

$$\varepsilon_{d,p} = \frac{\partial d}{\partial p} \frac{p_{\text{ref}}}{d_{\text{ref}}} = -\frac{1}{b_d} \frac{p_{\text{ref}}}{d_{\text{ref}}}, \quad (2.27)$$

Thus, a linear demand curve can be designed to imply a point elasticity at a reference point where a_d and b_d can be found by:

$$b_d = -\frac{p_{\text{ref}}}{\varepsilon \cdot d_{\text{ref}}} \quad (2.28)$$

$$a_d = p_{\text{ref}} + b_d \cdot d_{\text{ref}} \quad (2.29)$$

2.4.2 Piecewise linear demand approximation

In a piece-wise linear demand curve, the curve is simply split into several linear segments forming a curve. This can be done to approximate a log-log demand curve with constant elasticity (like in [19]) or to represent different utility functions for varying quantities of demand. The latter is the case for this thesis.

The segmentation consists of various segments where each segment is modelled through its own load shedding generator with the quadratic cost function as presented in Equation 2.23:

$$C(g_i) = (a_{d,i} - b_{d,i} \cdot D) \cdot g_i + \frac{b_{d,i}}{2} \cdot g_i^2 \quad \forall i \in \mathcal{F}$$

Where $\mathcal{F}$ is the set of flexible load shedding generators and the parameters $a_{d,i}$ and $b_{d,i}$ are the coefficients from Equation 2.28 and Equation 2.29 specifying the linear demand curve of the segment. The specific parametrisation of the segments is present in subsection 3.4.1.

For a more detailed proof of the PWL-segmentation, I refer to Appendix A in the paper by Brown, Neumann, and Riepin [19].

Quadratic problem

This modification makes the problem quadratic and non-linear. As discussed in 2.2.1, linearity is a sufficient condition for convexity, and convexity is important for the interpretation of the results. The condition under which a quadratic optimisation problem can be convex is if the quadratic term is concave for maximisation, or convex for minimisation [26]. This is determined by the sign of the constant before the quadratic term. As our problem is a minimisation problem, the sign must be positive in order to be convex. As seen in earlier equation 2.23:

$$C(g) = a_{\text{gen}} \cdot g + b_{\text{gen}} \cdot g^2$$

That is exactly the case for b_{gen} , meaning that convexity holds for our demand flexibility modelling and thus the optimal solution found is guaranteed to be a global optimum.

2.5 Model input data and preprocessing

The model's input data consist of hourly time series data for demand and weather. For each generation and storage technology, a set of techno economic parameters is also given.

2.5.1 Time series data

The used weather time series of wind and solar has already been converted to hourly capacity factors by Victoria and Andresen [28, 29].

A capacity factor (CF) is a number between 0-1 and a representation of the ratio between the actual production and the rated power/installed capacity of a generator. By having an hourly CF, the ratio expresses the weather conditions at the current hour through the power production of the generator. The formula used is:

$$CF_h = \frac{P_h}{P^{\text{rated}}}, \quad (2.30)$$

where P_h is the actual power output in hour h and P^{rated} is the installed (rated) capacity of the technology. That way, the actual generation can be found by multiplying by the models chosen installed capacity: $p_{\text{nom}} \cdot CF$.

Then the only other preprocessing of the solar and wind data consists of removing the data for 29th February in the leap years to be able to consistently plot and compare the time series results from each model run. The same is done for the demand time series.

The demand time series consists of aggregated values at the country level, given in hourly time steps. The data is already preprocessed and prepared [30], and besides moving the leap year day, the data is kept as is. The used data is from the datasets column: "ES_load_forecast_entsoe_transparency"

2.5.2 Techno economic parameters and annuitization

A comprehensive collection of techno economic cost data is used as input to the PyPSA model and is taken from the publicly available PyPSA technology data repository [31]. The dataset compiles investment costs, fixed and variable operation and maintenance costs, fuel costs, efficiencies, lifetimes, and discount rates, etc., based on reports from energy agencies such as the Danish Energy Agency and IEA, industry sources, and scientific research papers.

To ensure unit consistency across all technologies, cost parameters expressed per kW are converted to a per MW basis. Since the optimisation model is solved on an annual basis, investment costs are annuitized such that capital expenditures are represented as equivalent yearly costs over the technical lifetime of each asset. This allows cost recovery to be treated consistently within the optimisation time frame of a year.

The annuity factor a is computed as

$$a = \frac{r}{1 - (1 + r)^{-n}} \quad (2.31)$$

where r is the discount rate and n is the asset lifetime in years. A discount rate of 7% is used for all technologies.

The annualised capital cost is then calculated as

$$c^{\text{capital}} = \left(a + \frac{\text{FOM}}{100} \right) \cdot c^{\text{inv}} \quad (2.32)$$

where FOM is the fixed operation and maintenance cost expressed as a percentage of the investment cost, and c^{inv} is the investment cost from the dataset given in EUR per MW.

Marginal generation costs c^{marginal} are computed as the sum of variable operation and maintenance costs c^{VOM} and fuel costs c^{fuel} adjusted by the energy conversion efficiency η :

$$c^{\text{marginal}} = c^{\text{VOM}} + \frac{c^{\text{fuel}}}{\eta} \quad (2.33)$$

2.6 Validation and testing philosophy

In this thesis, three core optimisation models are employed, and this section outlines the methodological principles behind how and why these models are compared and validated against each other.

A direct comparison of price outcomes between the capacity expansion (CE) model and the dispatch models can be difficult to interpret. This follows from the fundamental differences in price formation between the models. The CE model includes capital investment costs and enforces cost recovery through endogenous capacity expansion, whereas the dispatch-based models exclude investment decisions and form prices solely based on short-run marginal costs. As a result, prices from the CE model and the dispatch models have different economic interpretations, making it more difficult to compare the models.

Thus, to enable structured comparison of operational behaviour, installed generation capacities derived from the CE model are treated as fixed, exogenous inputs to the dispatch models. Mean installed capacities across all simulated years are used, rather than year-specific capacity outcomes. Methodologically, this avoids evaluating dispatch performance in a system that is perfectly optimised for each individual weather year, which would otherwise suppress scarcity events and create biases in the extreme event identification.

Hence, two types of dispatch models are considered, namely the perfect foresight model and the limited foresight model. While the CE model also assumes perfect foresight, its operation differs due to its endogenous investment decisions and cost recovery constraints. The perfect foresight dispatch model, therefore, serves as a reference case that shares the same operational structure as the limited foresight model, differing only in their foresight assumptions. This allows us to more clearly analyse the isolated effects of available foresight on system operation and extreme event identification. This setup of using the three model types is shown in Figure 2.2 to give a visual intuition of how these principles function. Although not shown in the figure, the dispatch models are also compared with the capacity expansion model.

Consequently, the validation in this thesis is performed internally through cross-model comparison, focusing on consistency and differences in operational outcomes under varying modelling assumptions rather than comparison with empirical observations. This approach is consistent with the objective of assessing how modelling assumptions influence extreme system event identification and the role of long-duration energy storage.

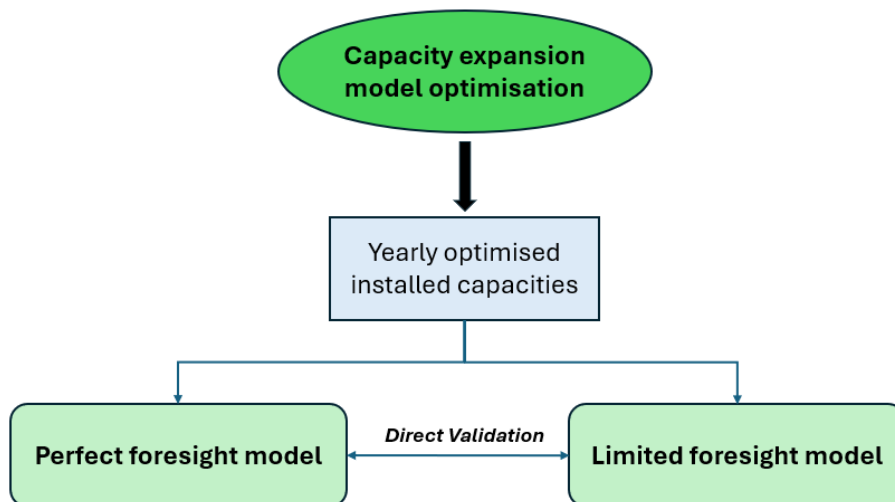


Figure 2.2: Basic model overview

2.7 Methodological limitations

The modelling approach adopted in this thesis is intentionally stylised in order to isolate the effects of limited foresight and demand flexibility on extreme system events. As a result, several methodological and modelling-specific limitations are implied, which should be considered when interpreting the results. These are therefore briefly listed below:

- The energy system is represented as a single-node, copper-plate model without transmission constraints. Spatial effects, such as including locational mismatches between energy production and consumption, and thus transmission line congestion, are therefore not captured.
- The analysis is restricted to the electricity sector and does not include sector coupling to heat, transport, or industry, thereby excluding potential sources of cross-sectoral flexibility.
- Simulations are based on historical weather and demand data and do not account for future climate conditions, electrification trends, or structural changes in electricity demand.
- Short-term operational constraints, such as ramping limitations, are not modelled. The operational focus is instead placed on intertemporal dynamics and multi-day system stress relevant for the LDES.

The implications of these limitations on the result interpretation and their relevance for future energy system planning are discussed further in Chapter 5.

3 Modelling setup and implementation

This chapter serves as a bridge between the methodology and the results. It primarily describes how the different models and scenarios are implemented and configured with specific parameters, although some results are also included as they informed part of the parametrisation.

3.1 Testing framework and data for the case study

3.1.1 PyPSA implementation

The model used in this thesis is PyPSA, which is short for Python for Power System Analysis and is an energy system model with various configurations and ways of representing energy systems as networks. It is possible to model different energy carriers, generation units, transmission links and lines, converters, etc. In this thesis, electricity is the primary energy carrier, whereas hydrogen serves as an intermediary energy carrier for producing and storing electricity. Below are listed the components that are relevant for modelling the system in this thesis:

- Bus
- Carrier
- Generator
- Store
- Link
- Load

The following sections explain how these components are connected to the data and the system modelled in this thesis. For a more detailed description of these components, see the model's documentation [32].

3.1.2 Model scope and case study selection

The model is a one-node model and has no transmission lines connecting it to other places/nodes. The model's geographical scope is Spain, chosen for its strong potential for renewable energy from wind, solar, and hydropower. In the capacity expansion optimisation, the model is doing green field optimisation planning, meaning that it assumes that there are no installed power generation capacities. Instead, the model can choose to optimise the system based on the following technologies:

- Onshore wind turbines
- Solar PV energy systems
- Hydrogen salt cavern storage
- Electrolysers to produce hydrogen
- Fuel cells for converting hydrogen into electricity

Initially, offshore wind turbines were also considered, but since the production pattern is similar to onshore wind, this technology was removed to keep the system simple and to have a single representation of wind energy.

3.1.3 Model input data

The specific numbers for the techno economic data used in this thesis are shown in Table 3.1. All the data is taken from PyPSA's technology database [31] for the year of 2030, although the CAPEX (capital expenditure) and marginal cost are calculated as described in section 2.5. All costs in the table are expressed in EUR 2020, and consequently, all monetary results presented in this thesis refer to this currency year.

Table 3.1: Cost parameters for selected technologies in the Spanish case study.

parameter units:	CAPEX EUR/MW/year	FOM [%/year]	VOM [EUR/MWh]	marginal cost [EUR/MWh]	efficiency []	lifetime [years]
Electrolyser	188,715 ²	4.00	0.00	0.00	0.62	25
Fuel Cell	223,935 ²	5.00	0.00	0.00	0.50	10
H ₂ cavern storage	149 ¹	0.00	0.00	0.00	1.00	100
Onshore Wind	101,644	1.22	1.43	1.43	1.00	30
Solar PV	51,346	1.95	0.01	0.01	1.00	40

1: in EUR/MWh, 2: in EUR/MW_{el}

The demand data used in the model's base configuration is that from the dataset Open Power System dataset [30] of the year 2018/2019. The weather data used is from the years 1979-2010 (Victoria and Andresen [28, 29]).

Modelling storage

In PyPSA, there are two main ways of modelling an energy storage, either as a *storage unit* or as a *store* with two *links*. The storage unit combines and connects the energy capacity with the power capacity through the input of duration time. In short, it means that only the power capacity is optimised, and then the energy capacity is found as the relation between power and duration time. The other approach is more fundamental and allows for the energy capacity to be optimised independently of the charging and discharging power of the storage, as opposed to the storage unit. I have chosen to model the hydrogen storage as a "store with links", allowing the model to optimise and size the electrolyser, fuel cell and cavern storage independently.

Cyclic SOC modelling

In PyPSA, a storage can be modelled with different configurations of initial SOC and behaviour. The energy stored at the start and end is important for the total energy system balance and for the optimisation of the problem. Depending on whether it ends with more or less energy than it started with, the storage will act as a net consumer or net generator (respectively). To avoid this, a cyclic condition (in PyPSA: *e_cyclic*) can be imposed to ensure that the storage will start and end the year with the exact same amount of energy. This cyclic approach is used to run the capacity expansion and economic dispatch model with perfect foresight. The case is slightly different for the limited foresight model using rolling horizon, which will be elaborated in section 3.3.

3.1.4 Experimental setup and test scenarios

The experimental setup consists of a base scenario with three types of models. As described in section 2.6, the study includes a capacity expansion model (CE), an economic dispatch model with perfect foresight (PF), and an economic dispatch model with limited foresight (LF). As an additional scenario of modelling flexible demand, the two dispatch models are extended to include elastic demand. Consequently, 5 models are run and compared with respect to event identification, price formation, and long-duration energy storage. A visualisation of this is shown in figure 3.1

To be consistent in their comparison, they all use the same yearly data, i.e. weather years, demand years, techno economic data and include the same technologies.

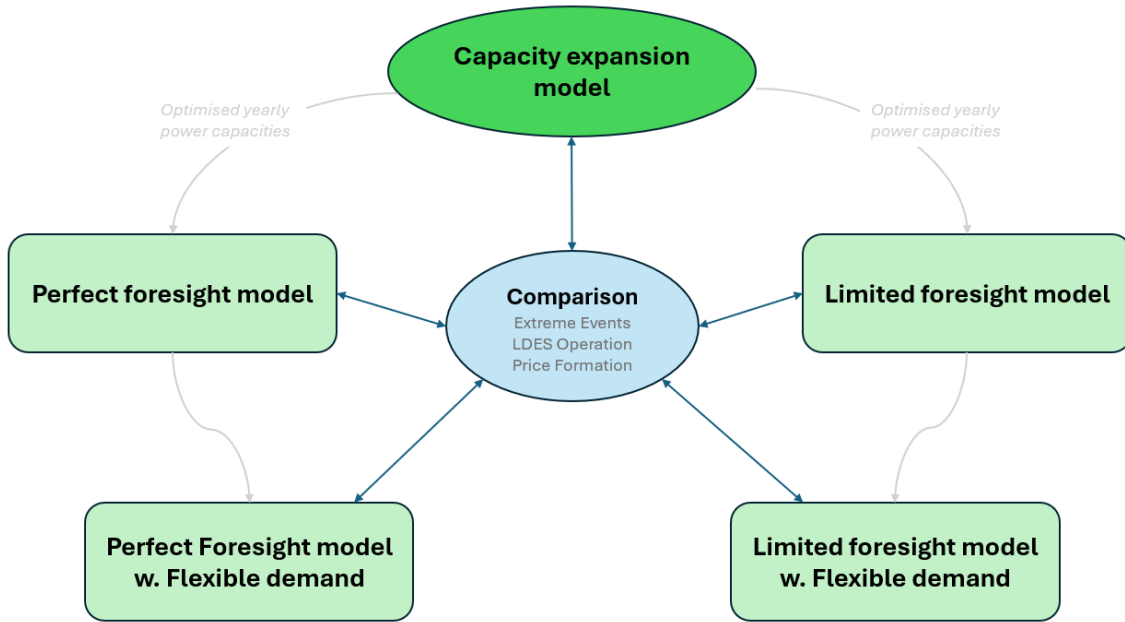


Figure 3.1: Overview of the 5 models

Software implementation and reproducibility

The entire analysis in this thesis is conducted in Python using PyPSA version 1.0.6. The code required to reproduce the analysis, results, and plots presented in the thesis is available in the GitHub repository *Master-Thesis-Jonathan-Gadfelt* [33]. The repository includes a "README" file with instructions for running the code.

3.2 Capacity expansion model

3.2.1 Formulation of the full model

In the subsection 2.1.2, the fundamental dynamics of the capacity expansion model were presented, but now I will introduce how the energy storage is modelled theoretically and thus present the full problem:

$$\min_{G_i, g_{i,t}, E_t, S^{\text{ch}}, S^{\text{dis}}, s_t^{\text{ch}}, s_t^{\text{dis}}} \sum_{t \in \mathcal{T}} c_i^{\text{inv}} G_i + \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} c_i^{\text{var}} g_{i,t} \quad (3.1)$$

$$\text{s.t.} \quad \sum_i g_{i,t} + s_t^{\text{dis}} - s_t^{\text{ch}} = D_t, \quad \forall t \in \mathcal{T} \quad (3.2)$$

$$0 \leq g_{i,t} \leq G_i, \quad \forall i, t \in \mathcal{I}, \mathcal{T} \quad (3.3)$$

$$E_t = E_{t-1} + \eta^{\text{ch}} s_t^{\text{ch}} - \frac{1}{\eta^{\text{dis}}} s_t^{\text{dis}}, \quad \forall t \in \mathcal{T} \quad (3.4)$$

$$0 \leq s_t^{\text{ch}} \leq S^{\text{ch}} \quad \forall t \in \mathcal{T}, \quad (3.5)$$

$$0 \leq s_t^{\text{dis}} \leq S^{\text{dis}} \quad \forall t \in \mathcal{T}, \quad (3.6)$$

$$0 \leq E_t \leq E^{\text{max}}, \quad \forall t \in \mathcal{T} \quad (3.7)$$

$$G, S, E, D, g, s, c \in \mathbb{R}^+ \quad (3.8)$$

The main behaviour of the storage is described in Equation 3.4, which defines the intertemporal behaviour of the storage state of charge E_t , and introduces temporal coupling through the inclusion of the SOC of the previous time step E_{t-1} . The storage efficiencies η^{ch} and η^{dis} are both unit-less decimal numbers between zero and one: $0 \leq \eta \leq 1$. The specific efficiencies are shown in Table 3.1.

The equations 3.7, 3.5, and 3.6 are all constraints limiting the storage's power and energy capacities. Namely by restricting the state of charge E_t and charging and discharging power ($s_t^{\text{ch}}, s_t^{\text{dis}}$) by the installed capacities $E^{\text{max}}, S^{\text{ch}}, S^{\text{dis}}$ which are endogenously defined by the capacity expansion problem.

3.2.2 Preliminary results of the capacity expansion model

The following sections present selected results from the capacity expansion model that were used to set up and parametrise the dispatch models.

Selecting the starting month and initial state of charge

Since extreme periods predominantly occur during the winter months, I allowed the model to simulate an entire, internally consistent winter season without distortions arising from an artificially elevated initial state of charge imposed mid-winter (on 1 January). To determine the optimal starting time, I examined the temporal evolution of storage's state of charge (SOC) over the full year, as obtained from the capacity expansion model when run for a normal calendar year. Figure 3.2 shows the average monthly SOC for all weather years, which provides a rough estimate of when the storage is generally at its highest.

The specific SOC levels were then analysed statistically, and 1st of June was selected to ensure that the storage and system were at a stable, high level. 1st of July was also considered, but because that would mean cutting the summer in half, June was selected over the two options.

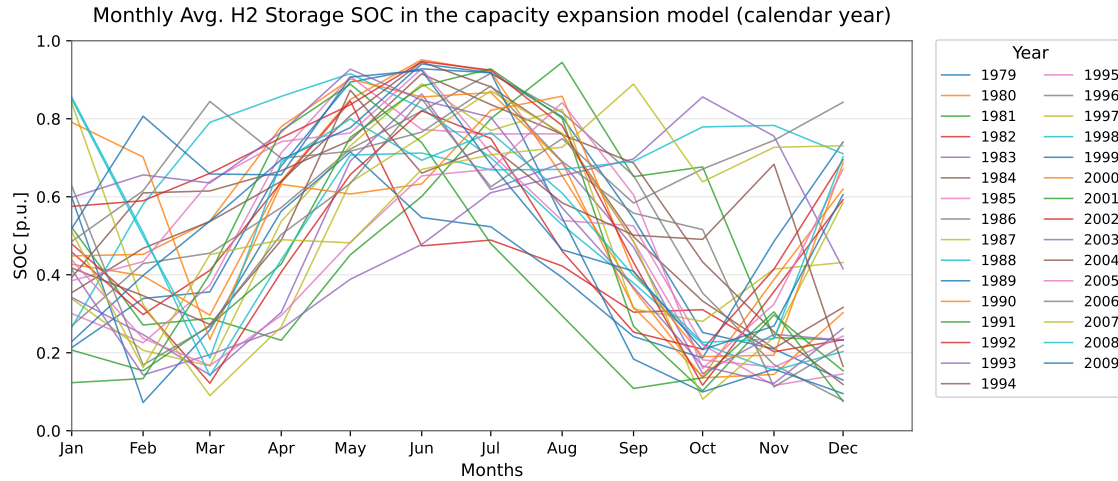


Figure 3.2: Monthly average SOC from the capacity expansion models initial runs using normal calendar year (jan-dec)

Capacity results

The models' optimised power capacities are relatively consistent throughout the year, as can be seen visually in Figure 3.3. It is noticeable that the capacities of the fuel cell are multiple times greater than those of the electrolyser. The interpretation is that the model prefers to have a higher discharger power, such that the model can supply the system in times of low VRE production and then slowly build up the energy in the storage when the production is plentiful.

Of particular interest is the comparison with the mean and maximum demand, which clearly illustrates that the system's nominal capacities must be substantially higher to accommodate periods during which both wind and solar generation are very low, even though the fuel cell capacity is approximately equal to the mean demand. This emphasises the challenge of building a robust system solely based on renewables and shows the important role of having a large and less weather-dependent dispatchable capacity in the form of the fuel cell to discharge the LDES.

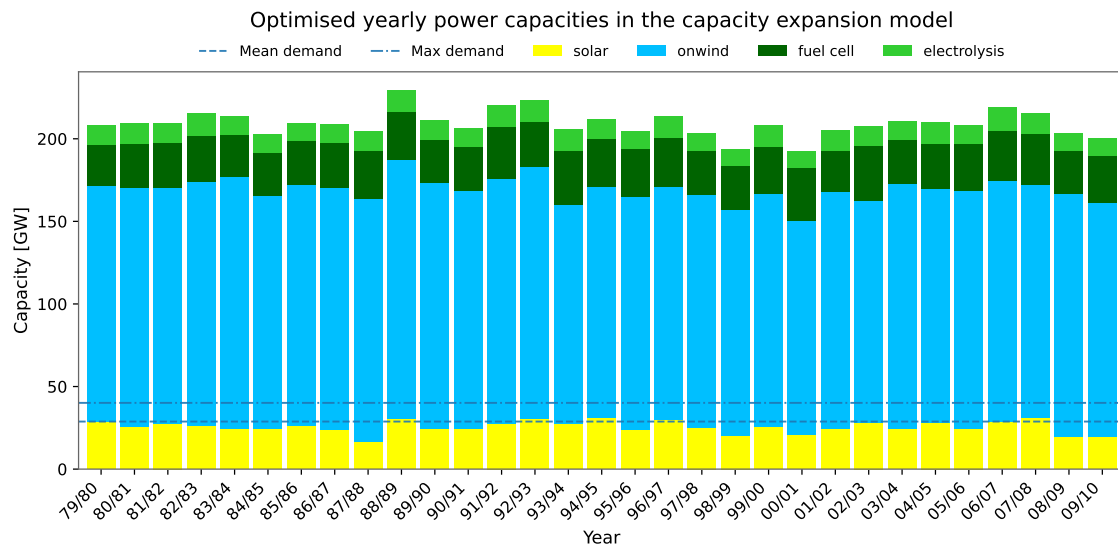


Figure 3.3: The optimised capacity values corresponding to each individual optimisation year derived from the capacity expansion model. Plotted with the mean and maximum demand

To further show the variation in capacities, Table 3.2 shows some basic statistical features of the capacities, such as the mean, various quantiles and standard deviation. The capacity of the hydro-

gen storage is also included here. With a minimum of 3.3 TWh and a maximum of 15.6 TWh, the energy capacity varies substantially. The specific values for all the yearly optimised capacities are shown in Table A.1 in appendix A.1.

Table 3.2: Optimal capacity statistics for capacity expansion model across the 31 years

	Onshore Wind	Solar PV	Electrolyser	Fuel Cell ¹	Hydrogen storage
unit	GW	GW	GW	GW	TWh
mean	143.59	25.64	11.91	56.26	9.47
std	5.63	3.65	1.10	4.23	2.68
min	129.48	16.28	9.72	49.89	3.29
25%	141.18	24.35	11.06	53.24	7.70
35%	141.36	24.62	11.40	54.08	8.95
50%	143.40	25.80	11.91	55.23	9.53
75%	147.28	27.91	12.94	58.09	11.32
max	156.36	31.15	13.97	65.58	15.64

1: The fuel cell capacities are for converting hydrogen. Thus, the electricity capacity is half of the shown capacity

Cost recovery

The cost recovery is a feature of long-run equilibrium as discussed in chapter 2 and is showing that the system is in balance and that social welfare is maximised. Cost recovery is true when the total profit of a generator is 0. The profit is found by summing the capital cost with the variable cost (marginal cost times generation dispatch) and revenue (the shadow price λ_t times generator dispatch):

$$\text{profit} = \text{revenue} - (\text{variable cost} + \text{capital cost}) \quad (3.9)$$

$$\Rightarrow \text{profit} = \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} \lambda_t \cdot g_{i,t} - (c_i^{\text{var}} \cdot g_{i,t} + c_i^{\text{inv}} \cdot G_i) \quad (3.10)$$

This is done for all the generators and is shown in Table 3.3 for a random year of 1995. This also holds for all the other optimised years.

Table 3.3: Cost recovery for all components in the capacity expansion model for weather year 1995

Technology	revenue	variable cost	capital cost	total cost	profit
Onshore Wind	14,726.5	366.8	14,359.7	14,726.5	0.0
Solar PV	1,223.2	0.4	1,222.8	1,223.2	0.0
Electrolyser	2,680.6	633.3	2,047.3	2,680.6	0.0
Fuel Cell	10,240.0	3,841.3	6,398.6	10,240.0	0.0
Hydrogen storage	3,841.3	2,680.6	1,160.8	3,841.3	0.0

All values are given in million euros [MEUR]

Objective and mean prices

For reference, the objective value, mean shadow price, and mean marginal storage value (dual variable of the storage constraint) are presented in Table A.2 in the appendix. The mean across the years of the latter two will be used to set up the dispatch model and will be discussed further in the next section.

3.3 The economic dispatch models

3.3.1 Perfect foresight model

The perfect foresight model used in this study is similar to the economic dispatch model presented in subsection 2.1.2, with the difference of having an energy storage modelled. Compared to the capacity expansion problem (3.2.2), the only difference is the exclusion of the installed capacities G_i , S^{ch} , S^{dis} , E_t as variables and thus also the removal of the investment cost c^{inv} in the objective function. The full problem is kept in subsection A.1.3 for reference.

3.3.2 Setup in the two models

Capacities used in the model

The dispatch model uses the mean of the optimised capacities as shown in Figure 3.4.

Installed power capacity in the dispatch models [GW]

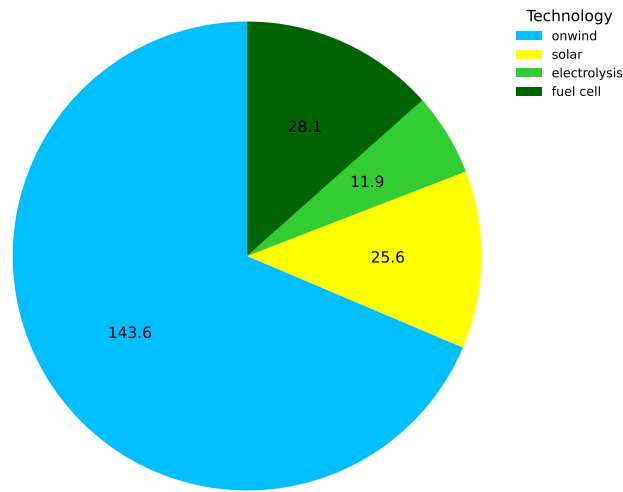


Figure 3.4: Power capacity mix used for all the dispatch models.

The energy storage capacity of the underground hydrogen storage is also set to the mean value from the capacity expansion model results, namely 9.47 TWh. (Table 3.2).

The dispatch model is implemented in PyPSA by fixing all generation and storage capacities and optimising only their operation. In PyPSA, the nominal capacity parameter p_nom is set equal to the optimal generation and storage power capacity variables G_i , S^{ch} , and S^{dis} . The values of p_nom are taken as the mean capacities obtained from the capacity expansion results. To prevent any investment decisions, all capital cost parameters c_i^{inv} are set to zero, and capacity expansion is disabled by setting $p_nom_extendable$ to False. As a result, the optimisation minimises total system cost based solely on marginal operating costs subject to the fixed capacity constraints. The implementation details are available in the *Build_dispatch_network* class in the file *Classes* in the thesis' associated GitHub repository [33].

Load shedding generator

Given that the mean capacities are used, the system will not be able to find a feasible solution for all years. To allow for an optimal solution to be found, a load-shedding generator is added to the models. This load shedding generator is created without any power limitations ($p_nom_extendable = \text{True}$), such that the model can freely decide the capacity needed for a feasible solution.

The estimated price of load shedding or VoLL (value of lost load) varies across countries, consumer sectors, and methodological approaches [34, 35]. Schröder and Kuckshinrichs [34] review several

studies, methods and countries to find that the value can vary from a few thousand EUR/MWh to beyond 250,000 EUR/MWh. Although they find that the VoLL of private consumers generally lies between 10-25,000 EUR/MWh when studies use macroeconomic approaches and at a maximum of 10,000 EUR/MWh using a willingness-to-pay approach. Furthermore, in an analysis of VoLL price estimations across Europe from 2018 [35], the price is set to vary between 1,500-22,940 with a median across Western Europe of 11,010 EUR/MWh. Thus, it is chosen to use 10,000 EUR/MWh as the VoLL, i.e. the marginal price of the load shedding generator. Although this value may not be consistent for all consumer groups, industry, commerce, public, etc., the main point of modelling VoLL in this study is to make it expensive and undesirable for the system to choose this as an option, such that it will only do it as a last resort.

Limited foresight dispatch model

The limited foresight model is implemented by applying the rolling horizon optimisation function from PyPSA when optimising the economic dispatch model (same models as used for the perfect foresight model). The function has two main inputs: *horizon* and *overlap*. The horizon is the time period for which the model can optimise in each window. For simplicity, this is set to 168 hours (7 days). The overlap determines whether the model can look back at certain snapshots of the previous optimisation window. This allows the model to consider the period around a window twice and achieve a smoother transition. However, given the simplicity of this model, no overlap is included.

Furthermore, as the models do not incorporate ramping constraints on generation or on charging and discharging power, and no other intertemporal variables are present, the storage state of charge E_t is the only variable that links decisions across the rolling-horizon windows.

3.3.3 Storage setting and operation

The storage configuration is where the models differ the most in their setup. As the rolling horizon function is essentially an algorithm for doing sequential optimisation as described earlier in subsubsection 2.1.2, it has some limitations regarding how it can be used. The initial analysis showed that the cyclic condition on the storage meant that energy would be artificially replenished to the storage between windows, as this condition would use the end SOC as the initial SOC. This means that the SOC sometimes changes drastically between windows, to either provide or remove substantial amounts of energy to the storage.

For these reasons, the limited foresight model is run without the cyclic condition. However, the perfect foresight model uses the cyclic condition. Without the cyclic condition, the limited-foresight model's initial SOC was 0, which contradicts the point made in subsubsection 3.2.2 about selecting a starting point where the storage is usually full. Thus, an initial SOC value is used for all the LF model runs.

Determining the value of the initial SOC

A statistical analysis of the system state at the start of the simulation period shows that, when the model is initialised on the 1st of June, only 13% of simulated years have an initial state of charge below 65%. Therefore, an initial SOC of 65% is chosen for the limited foresight model, as it provides starting conditions that are similar to those observed in the perfect foresight capacity expansion results, while avoiding to provide the LF model with additional energy in years with more challenging system conditions.

Marginal storage value

Another challenge with the operation of the storage with the rolling horizon configuration is that, as it does not consider the full year, it will not charge the storage unless needed within each specific optimisation window. An approach to tackle this is discussed and demonstrated by Brown, Neumann, and Riepin [19], who suggest that using the mean marginal storage value (MSV) from the capacity expansion model can be used to simulate how a storage operator would bid in the market, which means that the operation of the storage is then similar to that of the perfect foresight model.

The MSV is found from the Lagrangian dual variable of λ_t^{ES} , which is the KKT multiplier of the storage balance equation:

$$E_t = E_{t-1} + \eta^{\text{ch}} s_t^{\text{ch}} - \frac{1}{\eta^{\text{dis}}} s_t^{\text{dis}} \Leftrightarrow \lambda_t^{\text{ES}} \quad \forall t \in \mathcal{T} \quad (3.11)$$

The marginal storage value captures the intertemporal value of stored energy and is used to formulate storage bids in economic dispatch. As mentioned in chapter 2, the KKT multiplier of a constraint represents the marginal value of relaxing that constraint, which in this case gives λ_t^{ES} a natural interpretation as the opportunity cost of storing energy over time.

For the limited foresight model, the marginal storage value is defined to be the mean for all MSVs in the capacity expansion model(section A.1) at $\lambda^{\text{ES}} = 85.5$ EUR/MWh. With a charging efficiency of $\eta^{\text{ch}} = 0.62$ and a discharging efficiency of $\eta^{\text{dis}} = 0.5$, this results in an effective bid price of $85.5 \cdot 0.62 = 53.16$ EUR/MWh when charging, and $85.5/0.5 = 171.0$ EUR/MWh when discharging.

Modelling note: Effects of different foresight on system costs and price formation

Although not central to the main analysis, I briefly remark on a counter-intuitive observation made during the initial stages of this study, as it partly informs the interpretation of the results.

By construction, the perfect foresight (PF) dispatch model can operate storage more optimally than the limited foresight (LF) rolling horizon model. The natural expectation is therefore that PF should result in lower system costs, and indeed, the PF model consistently yields lower load shedding volumes. However, for several of the weather years, the total cost and objective value are higher in the PF model than in the LF model.

This behaviour is primarily explained by differences in price formation driven by storage operations. In the dispatch models, capital costs are excluded, and prices reflect short-run marginal system costs rather than long-run cost recovery, like in the CE model. Under PF, storage operation is fully intertemporal across the entire year and internalises future scarcity through bidding with its opportunity cost. As a result, future load shedding can propagate backwards in time, leading to price spikes even in hours without active load shedding. The corresponding price duration curves presented later in Figure 4.2 illustrate this effect.

In contrast, storage operation in the LF model is limited to the rolling horizon window. And with a fixed marginal storage value of 85.5 EUR/MWh, the bidding of the storage led to a more stable price formation. Differences in objective values are further influenced by storage boundary conditions, as the LF model is not subject to a cyclic state-of-charge constraint and may end the year with a different state of charge than it started (Figure B.7). These counter-intuitive effects are not considered critical for addressing the main research questions, as extreme event identification is performed relative to prices and costs within each model and weather year (section 2.3). The focus of the analysis is on how modelling assumptions influence the identification of extreme events, rather than on interpreting and comparing absolute system costs across the models.

3.4 Implementation of demand flexibility

The modelling of flexible demand is only done for the dispatch models with perfect and limited foresight, creating two new dispatch models. The following section will build on the theory section 2.4 and specify the data used in the models.

Following the theoretical definition in section 2.4 of how flexible demand can be modelled as load shedding generator(s) with a quadratic cost function, I will now explain how this is implemented and define what values are used for the specific parameters.

3.4.1 Defining piece-wise linear segments and their elasticity

It is chosen to model the flexible demand using piece-wise linear (PWL) segments to form an aggregated demand curve. This is chosen rather than modelling a single linear, flexible demand curve, which would imply highly variable elasticities along the curve. The PWL approach allows control of the elasticities in each segment, allowing for a more detailed demand representation and is therefore preferred.

I have chosen to model three segments representing:

1. A demand group with a low elasticity for higher prices above 1000 EUR/MWh up until the point of load shedding at 10k.
2. A very flexible demand group with high elasticity around price 1000 EUR/MWh.
3. A slightly flexible demand at lower prices, starting at the mean price of 98 EUR/MWh from the CE model.

Segment 1 represents a slightly elastic and gradual flexibility across consumer segments as prices rise up until the point of load shedding. Segment 2 comprises flexible, large industrial consumers for whom electricity prices play a significant role in their marginal production costs. Examples could be steel and aluminium production. Segment 3 represents a slightly flexible segment at lower prices, which could be electric-vehicle charging for flexible private consumers or for larger aggregators.

To inform the choice of segment-specific elasticities, values reported in the literature on electricity demand elasticity were reviewed. While no studies were identified that directly provide elasticity estimates for the specific demand segments considered here, several empirical results provide indicative ranges that guided the calibration.

For residential consumers exposed to real-time pricing, Fabra et al. [36] estimate elasticities in the range of -0.0072 to -0.054 using data from the Spanish electricity market. At a more aggregated level, Scarazzato [37] report an overall demand elasticity of approximately -0.1. Using historical day-ahead prices from Germany and Austria, Bönte et al. [38] estimate a substantially higher elasticity of -0.43 for the period 2010–2014. Finally, a recent meta-analysis of electricity price elasticities by Sikl et al. [27] reports a mean elasticity estimate for Europe of -0.125, with a confidence interval ranging from -0.26 to 0.01.

Based on these ranges and the qualitative interpretation of the demand segments, the elasticities are set to:

- Segment 1: -0.025
- Segment 2: -0.3
- Segment 3: -0.01

The segments were defined such that segments 2 and 3 account for 10% of the yearly mean demand, while segment 1 covers the remainder of the demand up to the load-shedding price corresponding to 11.5% of the mean demand. In the following, I briefly describe how the elasticities are used to define the cost function of the load shedding generators.

3.4.2 Calibrating the segment price levels

To ensure that the aggregated demand curve is still continuous and the segments are not overlapping, I applied an iterative calibration of the parameters in each segment. The resulting aggregated

inverse demand curve is shown in Figure 3.5.

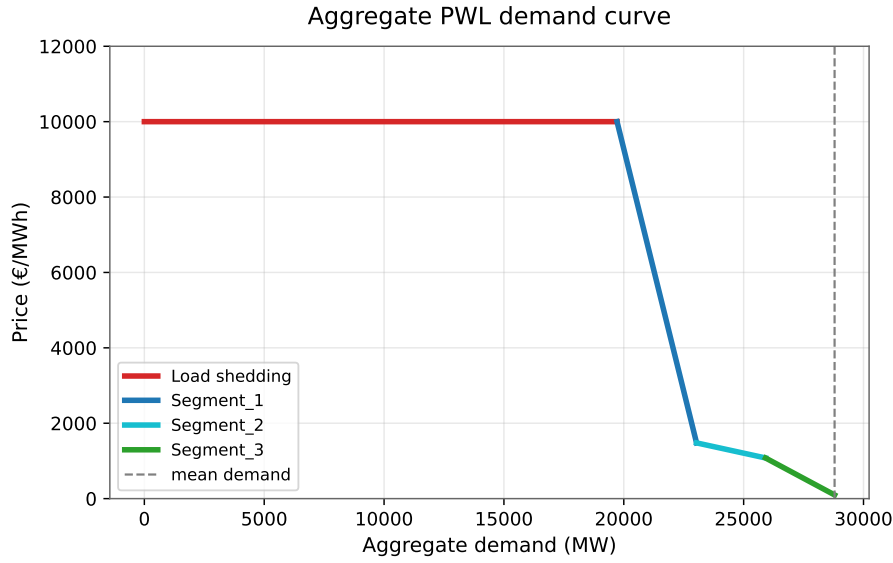


Figure 3.5: Aggregated inverse demand curve from using a piece-wise linear demand utility representation

The PWL elastic demand representation is calibrated around the mean system demand D , with each segment i associated with a reference demand level D_i^{ref} corresponding to a fixed share of D . Each segment is described by a linear inverse demand function of the form introduced in section 2.4,

$$p(D) = a_{d,i} - b_{d,i}D,$$

where $(a_{d,i}, b_{d,i})$ are the segment-specific coefficients.

Along a linear demand curve, the price elasticity of demand is not constant but varies with the operating point. The elasticities implemented in the model therefore correspond to *point elasticities*, evaluated at the reference point $(D_i^{\text{ref}}, p_i^{\text{ref}})$. The point elasticity, as defined in eqs. (2.25) to (2.27), is given by

$$\varepsilon_i = \frac{\partial D}{\partial p} \frac{p_i^{\text{ref}}}{D_i^{\text{ref}}} = -\frac{1}{b_{d,i}} \frac{p_i^{\text{ref}}}{D_i^{\text{ref}}}.$$

For the lowest-price elastic segment (segment 3), the reference price p_3^{ref} is anchored directly to the mean shadow price obtained from the capacity expansion model across all years, which was found to be 98 EUR/MWh. Together with the specified point elasticity ε_3 and reference demand D_3^{ref} , this uniquely defines the slope and intercept of the segment's inverse demand curve.

Starting from this low-price anchor, the inverse demand curve of segment 3 is "extended" towards higher prices by increasing the demand until the segment reaches its prescribed demand extent, which is given by the difference between the mean demand and the segment's reference demand. For segment 3, this is:

$$D - (D \cdot 0.1) = 28,798 - 2,879 = 25,914 \text{ MW} \quad (3.12)$$

The price corresponding to this demand level then defines the reference price for the subsequent segment.

This new price–demand point describing the end of segment 3 then serves as the reference point $(D_2^{\text{ref}}, p_2^{\text{ref}})$ for calibrating segment 2 using its specified point elasticity. The same procedure is repeated to define the reference point of the next segments.

Finally, segment 1 is extended until the load shedding price of 10 000 EUR/MWh is reached. At this point, the demand is then completely elastic at the high load shedding cost. The resulting parameters $(a_{d,i}, b_{d,i})$ are used to calculate the cost function of the generator coefficients as presented earlier in Equation 2.24:

$$a_{\text{gen}} = (a_d - b_d D) \quad \text{and} \quad b_{\text{gen}} = \frac{1}{2} b_d$$

The resulting parameters for each segment and generator are shown in Table 3.4.

Table 3.4: Calibrated parameters for elastic demand generators, evaluated at mean demand of 28.8 GW.

Segment	Demand share [%]	Quantity [GW]	ε	a_{gen} [€/MWh]	b_{gen} [€/MWh]
Segment 1	11.5	3.31	-0.025	1479.6	1.2847
Segment 2	10.0	2.88	-0.300	1079.0	0.0694
Segment 3	10.0	2.88	-0.010	98.0	0.1702

3.5 Parameters in the extreme event methods

3.5.1 Shadow price method

The so-called shadow price method is the one method outlined in section 2.3.1 where the model's cost, calculated by taking the shadow prices λ_t times the demand D_t , is used to determine system defining events. Recalling equation 2.16 that decides when an event occurs:

$$\sum_{t=t_0}^{t_0+T^{SP}-1} D_t \cdot \lambda_t \geq C$$

I now define the parameters C and T^{SP} and describe their role in the analysis.

The threshold C denotes a monetary value representing the maximum allowable cost that may be incurred over the time interval from t_0 to T^{SP} . If the system were well-known, representing a real system, the value could be set based on historical data. In contrast, for a more arbitrarily defined system and model, such as the one used for this thesis, a fixed number is hard to define as meaningful and comparable between the CE and dispatch models. Therefore, it is chosen to use a percentage of the total system cost instead, which is denoted as β such that $C^\beta = \beta \cdot C^{tot}$ where C^{tot} is the total system cost of the entire year. This now gives:

$$\sum_{t=t_0}^{t_0+T^{SP}-1} D_t \cdot \lambda_t \geq C^\beta$$

Where:

$$C^\beta = \beta \cdot C^{tot}$$

and:

$$C^{tot} = \sum_{t \in \mathcal{T}} D_t \cdot \lambda_t$$

This is consistently what is done in the entire analysis, which makes the comparison between different years and models more straightforward. Especially because the price difference coming from including investment cost in a capacity expansion model vs excluding them in the dispatch models would make it hard to compare the events from setting a fixed monetary number as the threshold C . Later in chapter 4, the analysis of varying this threshold is presented.

The minimum event duration, t_0 , is set to 2 days (48 hours) because the objective is to identify prolonged extreme system events. The maximum duration length T^{SP} is set to be 14 days in order to also capture longer renewable energy droughts. This is also consistent with what is done in [9], where the method was first introduced.

The detection algorithm

The detection algorithm used in this thesis follows the methodology proposed by Grochowicz, Greevenbroek, and Bloomfield [9] and is implemented as provided in the accompanying code base[39] in the file *difficult_periods*. The only changes made are removing the nodal summations, given that this study only includes one node.

The algorithm first computes an hourly system cost time series by multiplying the marginal electricity price at the system bus by the corresponding demand. All candidate intervals with durations between the minimum event length t_0 and the maximum super period length T^{SP} are then evaluated by computing their cumulative system cost. Intervals whose cumulative cost exceeds the threshold C^β are retained as candidates. To avoid double-counting, overlapping candidate intervals are resolved by selecting periods in descending order of cumulative cost and discarding any candidates that overlap previously selected events. Each selected event is subsequently expanded to include adjacent hours for which the hourly system cost exceeds the average hourly cost of the event. The final set of detected events may contain multiple non-overlapping periods, each of which is characterised by its start and end time and a peak hour, defined as the hour with the highest system cost within the event [39].

3.5.2 Net load event detection identification

Recalling that the net load L_t^{net} is defined as the time series representing electrical demand minus VRE production (see eq. (2.15)), I now describe the simple algorithm by which this concept is operationalised as an extreme event within the context of this thesis.

To identify net load events, a dynamic rolling window approach is used to determine the most severe period within each weather year. The net load time series is aggregated over a fixed-length rolling window, and the window with the highest cumulative net load is selected as the event period. The rolling window is evaluated for all possible starting times, ensuring that all candidate periods are considered. This approach ensures that only the most extreme period per year is retained, while competing and overlapping candidate periods are implicitly discarded through the maximisation over rolling windows. The identified event is characterised in the same way as for the shadow price method, namely by its start and end time and a peak hour. The peak hour is defined as the hour with the highest net load within the selected window. This algorithm shares similarities with the shadow price method, with the key distinction that it uses a fixed event duration of 7 days and does not allow multiple events per year.

3.5.3 Load shedding event identification

Load shedding events are identified using a rolling window approach similar to that used for net load events, but with dispatched load shedding power g_t^{ls} as the defining metric. To exclude short and negligible instances of load shedding, an aggregate threshold α^{ls} is applied over a fixed event duration T^{ls} , such that an event is identified only if:

$$\sum_t^{T^{\text{ls}}} g_t^{\text{ls}} \geq \alpha^{\text{ls}} \cdot \sum_t^{T^{\text{ls}}} D_t \quad (3.13)$$

The event duration T^{ls} is fixed to 7 days, while the threshold α^{ls} is defined dynamically as 0.25% of the total electricity demand within the event window. Although this value is relatively small, it could be increased if the analysed system is subject to frequent load shedding. In this study, however, the primary objective is to capture all time intervals characterised by a substantial amount of load shedding. The preliminary analysis indicated that events were sometimes identified on the basis of as little as one hour of load shedding. Consequently, the threshold parameter α^{ls} was introduced to mitigate this issue.

3.5.4 Flexibility activation event

This event type can not necessarily be classified as an extreme event, nor is it established in the existing literature. Instead, it is introduced here as a constructed category designed to facilitate a consistent comparison of the peak utilisation of flexible demand with the other event types.

Its definition closely resembles that of the load-shedding event. However, in this case, the metric being accumulated is the aggregated flexible demand from the three elastic load-shedding generators, as expressed in eq. (3.14). The event identification procedure employs the same dynamic rolling-window algorithm used for both net load and load shedding events.

$$\sum_i^{\mathcal{I}^{\text{flex}}} \sum_t^{T^{\text{flex}}} g_{i,t}^{\text{flex}} \geq \alpha^{\text{flex}} \cdot \sum_t^{T^{\text{flex}}} D_t. \quad (3.14)$$

Here, $\mathcal{I}^{\text{flex}}$ denotes the set of the three flexible generators and their flexible demand "activation" $g_{i,t}^{\text{flex}}$. T^{flex} represents the duration of the event, which is fixed at 7 days. The minimum required activated flexible demand is given by the threshold α^{flex} , defined as 3% of the total load over the event window. This threshold is selected to be marginally higher than the level employed for load shedding events, thereby reflecting the comparatively lower system cost of activating elastic demand and, consequently, its greater probability of being triggered at relatively low price levels, when the power system is not operating under stress conditions.

3.6 Final setup used for the results

This section summarises the final parameter values and modelling choices used to generate the results presented in Chapters 5 and 6.

- **Weather years:** 1979–2010, simulated from 1st of June to 31st of May for all years.
- **Demand data:** 2018–2019, from 1st of June to 31st of May. Total demand for Spain.
- **Installed capacities in dispatch models (fixed to mean CE values):**
 - Onshore wind: 143.6 GW
 - Solar PV: 25.6 GW
 - Electrolysis: 11.9 GW
 - Fuel cell: 56.3 GW
 - Hydrogen storage (energy capacity): 9.5 TWh
- **Load shedding cost (dispatch models):** 10,000 EUR/MWh
- **Limited foresight dispatch settings:**
 - Marginal storage cost: 85.5 EUR/MWh (mean marginal storage value from the CE model)
 - Initial storage state of charge: 65%
 - Rolling horizon length: 168 hours (7 days)
 - Horizon overlap: 0 hours
- **Elastic demand representation (dispatch models):**
 - Piecewise linear demand with three demand segments

- Mean system demand: 28.8 GW
- Segment capacities, elasticities, and calibrated cost parameters are reported in Table 3.4
- **Piece-wise linear parameters for the dispatch models with flexible demand:**
 - **Segment 1:** Power capacity (p_{nom}) 3.31 GW, marginal cost 1479.6 EUR/MWh, marginal quadratic cost 1.2847 EUR/MWh
 - **Segment 2:** Power capacity (p_{nom}) 2.88 GW, marginal cost 1079.0 EUR/MWh, marginal quadratic cost 0.0694 EUR/MWh
 - **Segment 3:** Power capacity (p_{nom}) 2.88 GW, marginal cost 98.0 EUR/MWh, marginal quadratic cost 0.1702 EUR/MWh
- **Extreme event identification parameters:**
 - Shadow price (SP) method: $\beta = 20\%$, duration 2–14 days
 - Net load method: duration 7 days
 - Load shedding method: $\alpha^{LS} = 0.25\%$, duration 7 days
 - Flexibility activation method: $\alpha^{flex} = 3\%$, duration 7 days

4 Results

4.1 General results

4.1.1 Price formation

Capacity expansion model

The prices in the CE model have the commonality that for each year, there is one hour with an extremely high price. The max price for multiple years is at approximately 224,000 EUR/MWh. This is a feature of the numeric solver trying to find a way to recover the cost of each generator. Cost recovery can occur slowly throughout the year or in a few hours with a very high price, where the latter is the case for one of the technologies in this model. Therefore Figure 4.1 show both the price duration curves logarithmic and for prices below 500 EUR/MWh. From the log. view in Figure 4.1a, it is clear to see how the extremely high price only occurs for a very short time (one hour) while the rest of the price formation is more dynamic. The model has prices near 0 for a very large part of the year (nearly half of the year), as the figures show. The exact lower price on the graphs is at 1.43 EUR/MWh, which is in this case the marginal cost of the wind generation. When looking at Figure 4.1b, it is clear how the price curves are formed stepwise, which can be explained by the simple nature of the system. Contrary to this, had there been more technologies or more nodes, this would have increased the variation of prices, which would have reduced the steps to smooth the curve a bit. Last takeaway from the price formation is how the graph clearly varies between the years but still follows the same pattern, and for most years seems to align around price 100 EUR/MWh, which aligns well with the mean price across all years being 98.7 EUR/MWh.

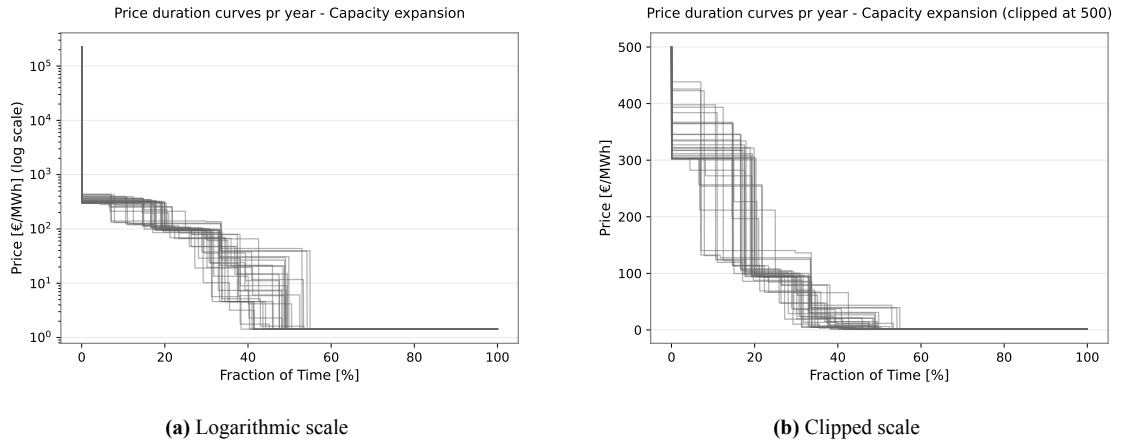


Figure 4.1: Price duration curves for the capacity expansion model

The economic dispatch models

Now, the price formation from the dispatch models is quite different. Looking at Figure 4.2a the price formation is a lot more uniform with less steps and variation. From a theoretical point of view, this makes sense as it is now only the operating cost of the generators that forms the generator's bids. An interesting thing is that the perfect foresight has more hours where the load shedding price of 10,000 occurs, as it was described earlier in 3.3.3. This is because the storage uses perfect foresight to set a high price, reflecting the opportunity cost of avoiding load shedding.

On the other hand, the limited foresight model, which is not restricted by a fixed amount of energy due to the absence of the cyclic condition, and only sees one week ahead, cannot use the storage to link the timesteps of scarcity hours throughout the year. Furthermore, it also has a fixed MSV for

the store, which means that for most hours, this value sets the prices, as can be seen in Figure 4.2b where the two biggest steps in the middle correspond to the price of charging and discharging the storage at 53.16 and 171.00 EUR/MWh, respectively.

Hence, one could argue that the high prices in the LF model are more guaranteed to represent a scarcity period of actual load shedding, whereas this is not always the case for the PF model.

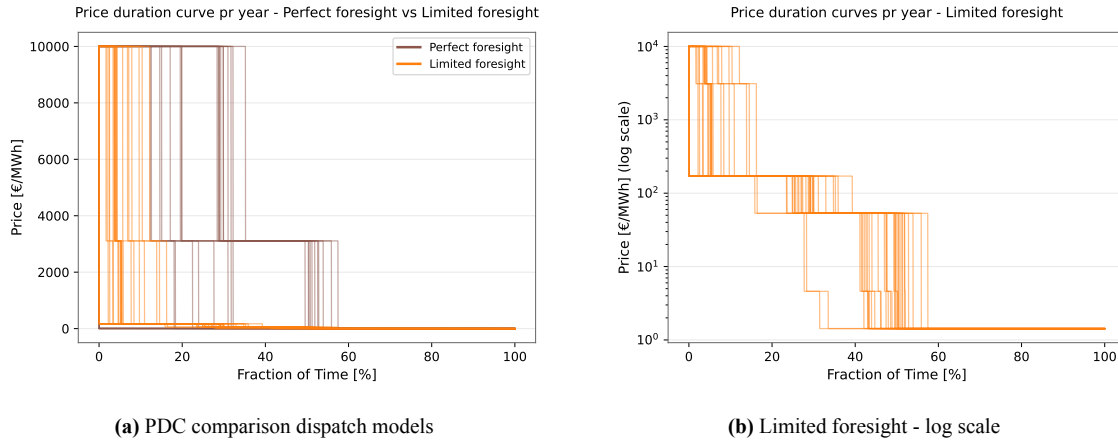


Figure 4.2: Price duration curves for the dispatch models with inflexible demand

The dispatch models with flexible demand

By applying flexible demand to the perfect foresight model, we see that it is actually able to completely avoid load shedding since it is able to fully utilise the flexible demand instead (Figure 4.4a). Although it has more yearly variation, there are still some bigger steps suggesting that the storage is again using the foresight to set a high price as an opportunity cost of using the flexible demand. In the curve between the steps, we see a slight smothering of the curve, showing the impact of the price elasticity from the piece-wise linear demand curve.

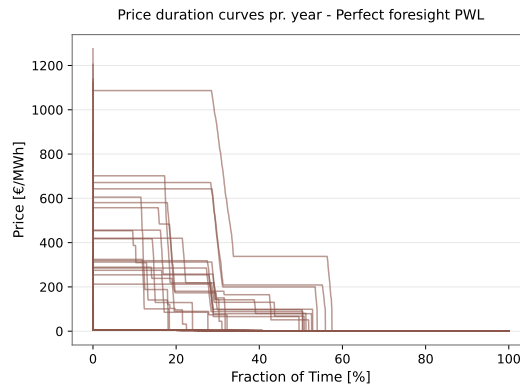


Figure 4.3: Price duration curves for the perfect foresight model with PWL-Flexible demand

The price duration curves of the limited foresight model with flexible demand is showed in Figure 4.4. In this model, we still see some load shedding occurring, however, reduced compared to Figure 4.2 and with many smaller steps along the curves. Nonetheless, the two big price steps seen in the base case are still present, as was also the case in fig. 4.4b.

4.1.2 Key metric comparison across models

Table 4.1 provides an overview of some key metrics describing important parts of the models and highlighting key differences. It is important to notice that these are averages across all 31 weather

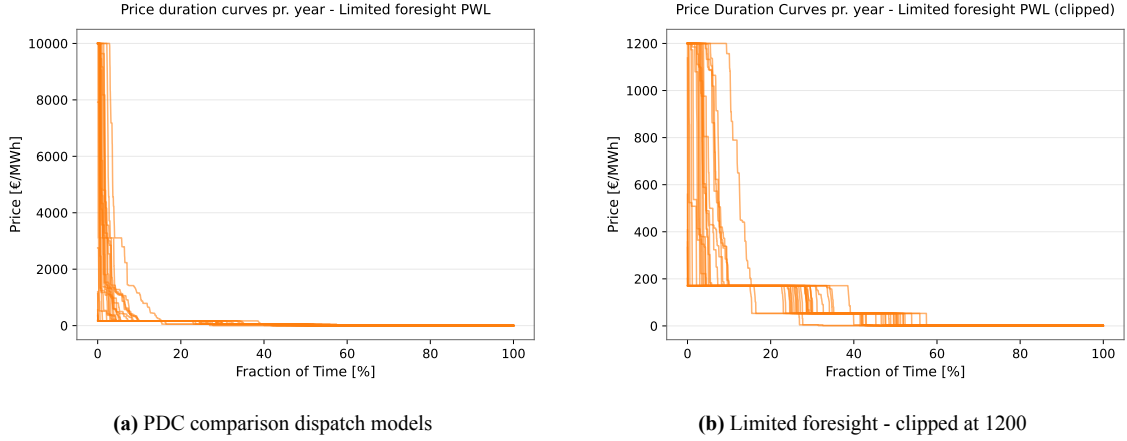


Figure 4.4: Price duration curves for the dispatch models with flexible demand

years, which is a rough simplification, but given the big difference between the 5 models, it helps show the bigger picture. The two metrics ENS and LOLH stand for loss of load hours and energy not served (respectively), where LOLH is a sum of how many hours load shedding occurs in, and ENS is how much energy is actually shed. Both are yearly sums which are then averaged across all years and shown in the table.

Interestingly, the mean price of the perfect foresight model being the highest among the models, shows what was also seen on the price duration curves. Additionally, we now see on the LOLH and ENS that even though less energy is "shed", it occurs over more hours, indicating that the PF can strategically stretch out LS over the years. For reference, the ENS in the LF model corresponds to 0.63% of the total demand, which is a substantial amount considering the model covers the entire Spanish system.

Looking at the unique prices in the models is a way to get an additional idea of how many steps there are in the model runs, which was also indicated by the steps shown by the price duration curves. It shows the simplicity of the models with the LF and PF base models only having an avg. of 3 and 5 different types of prices. Nevertheless, the number of prices increases significantly when elastic demand is implemented. Moreover, the ENS and LOLH drastically decrease for the LF model and even disappear completely from the PF model. Although it should be said that for the flexible models, some load is actually being "shed", but in the sense that it is elastic and flexible, this is not accounted as a loss in the same way.

Table 4.1: Comparison of average key metrics across the model

	Mean Price	Max Price	Unique Prices	ENS	LOLH	Max LOLH
<i>unit</i>	<i>€/MWh</i>	<i>€/MWh</i>	<i>[]</i>	<i>GWh/year</i>	<i>hours</i>	<i>hours</i>
Capacity Expansion	99	224374	8	0	0	0
Perfect Foresight	1820	10000	3	1374	222	1105
Limited Foresight	422	10000	5	1582	180	762
Perfect Foresight PWL	82	1276	93	0	0	0
Limited Foresight PWL	158	10000	97	147	40	250

ENS = Energy not Served, LOLH = Loss of Load Hours

4.2 Extreme event identification

In this section, I will present a graphical overview of how events are identified throughout the years and different models. The events are made with the default thresholds described in chapter 3. For

all the shadow price events, this corresponds to a value of 20% of the total system's cost occurring during the event.

4.2.1 The shadow price method across the models

A figure visually combining the events gathered using the shadow price (SP) event identification method in all models is shown in Figure 4.5. Only the months from September to April are shown since there are no events in the summer months (full figure in appendix B.1). The figure has a heatmap¹ background of the daily mean electricity prices gathered from the limited foresight model. Here, it can be seen that it always aligns well with all the events from the two LF models, but this is not the case for the PF and CE models. In the years 06/07, 92/93, and 88/89 the CE events align with those of the LF model, although the LF models have multiple events defined in those years. Generally, the CE model events are as short as the threshold t_0 allows (2 days) for all years. The reason is the very high price of the CE model, which means that 20% of the total cost falls within a 2-day period or less. There are a few events in the perfect foresight model, and they align closely with those in the CE model.

As it was seen with the price duration curves, there are more high prices in the PF model, and for that reason, the cost is actually spread more evenly out, which is why the method does not identify the periods at a maximum of 14 days. Additionally, some of the identified events are from years with a very short period of high prices, similar to the CE model. The SP event plot with PF shadow price heatmap is shown in the appendix B.2. In the few years where an event is defined, it is a few hours of load shedding that is causing this, since the rest of the year consists of very low prices from the wind and solar generation. Thus making a few hours of very high prices more significant compared to the total costs. Looking at the events from the two limited foresight models, we see how these are almost always occurring at the same time, but with significantly fewer and shorter events for the model with flexible demand.

Highlighting two of the key takeaways: The SP method works in a dispatch-only model with limited foresight, and there is a general tendency that events from the LF model occur earlier in the year compared to the CE model.

4.2.2 Comparing event identification methods in the limited foresight model

A key focus of this study is to analyse what happens with events during limited foresight, which is the focus of Figure 4.6, showing the 10 years from 1990-2000 in three subplots. The figure consists of the events in the limited foresight models with 3 different heatmaps as varying backgrounds for the same events. The upper plot is showing the daily mean shadow price (similar to fig. 4.5), the middle plot shows aggregated daily load shedding (the daily ENS), and the bottom plot shows the aggregated sum of the positive net load. All heatmaps are based on the outputs of the base limited foresight model (without flexible demand).

As can be seen in the figure, the events tend to identify the same extreme period across several years. Although we do see some years, 94/95 and 91/92, where the shadow price events disagree with the other methods, as can be seen from the heatmaps, the identified periods are indeed having prolonged periods of load shedding and high scarcity prices. An explanation could be that both the LS, NL and flex activation event method are restricted to identify one period, but given the indications from the heatmaps, we clearly see that these are scarcity periods.

Furthermore, the scarcity is confirmed on the net load heatmap in the bottom plot, where the two SP events in 94/95 occur in short periods of high net load. In general, there seems to be a quite clear overlap between periods of high load shedding and shadow prices as shown by the heatmaps. In contrast, the net load is high frequently throughout the year, and even though many net load events overlap with other events, the NL events do not seem completely robust for identification. An example is the net load event in 93/94 where we do not see high prices or load shedding, which

¹A heatmap is a visual color scale where colours change alongside values

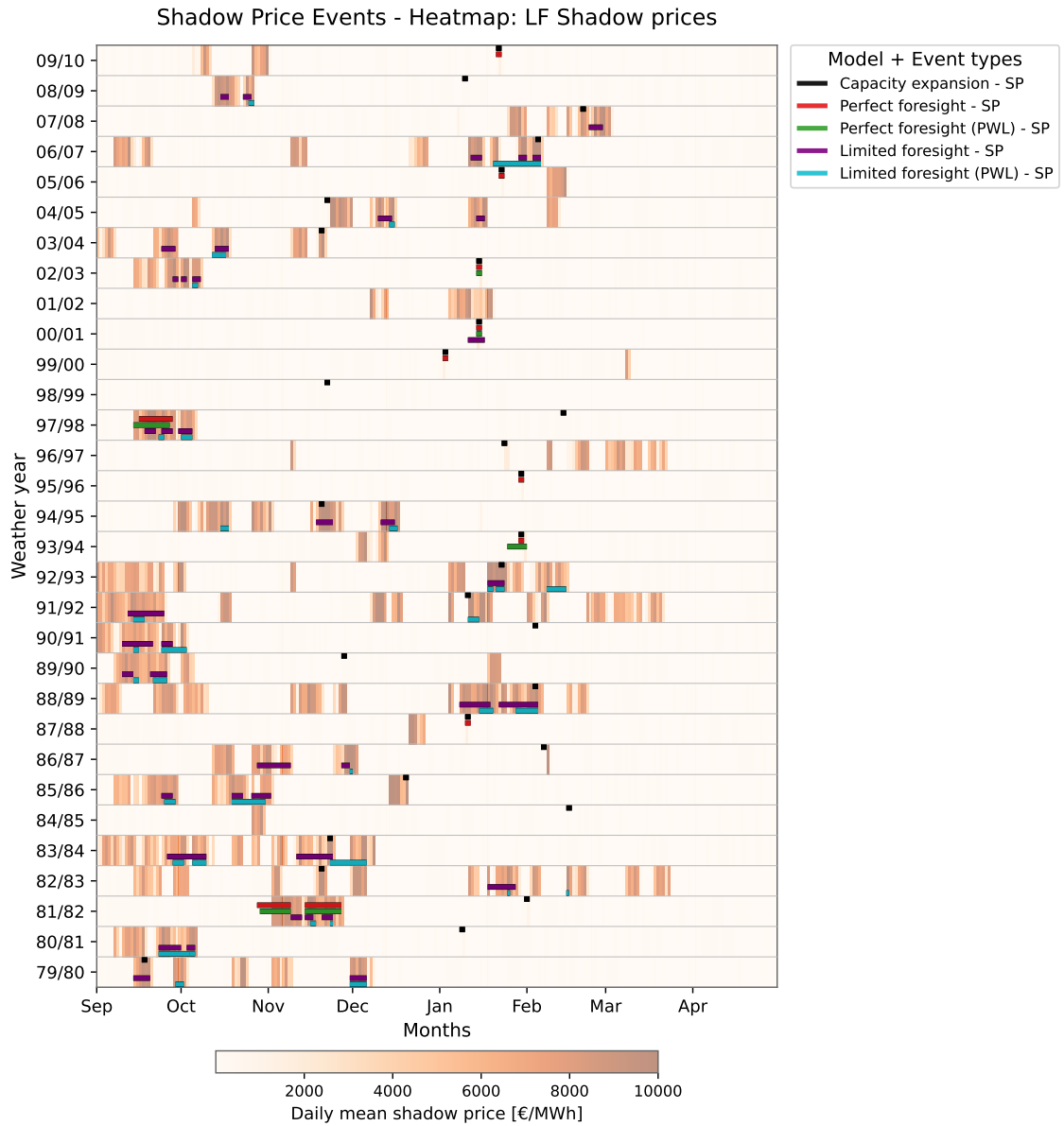


Figure 4.5: Identified shadow price events with heatmap background from the limited foresight model. Showing months from September to April.

Abbreviations explained: SP = Shadow price event, PWL = Piece-wise linear model, LF = limited foresight model.

is essentially the result of the storage (and partly the flexible demand) not being considered by the method. The net load generally varies a lot more throughout the year and might also be high at times during the summer months. To see more details of the annual net load pattern, a version of the figure showing the entire year can be seen in Figure B.4.

The event identification in the rest of the years, i.e. 2001-2010 and 1979-1989, follows the same patterns as described for 4.6. To view the full figures, see Figure B.5 and Figure B.6.

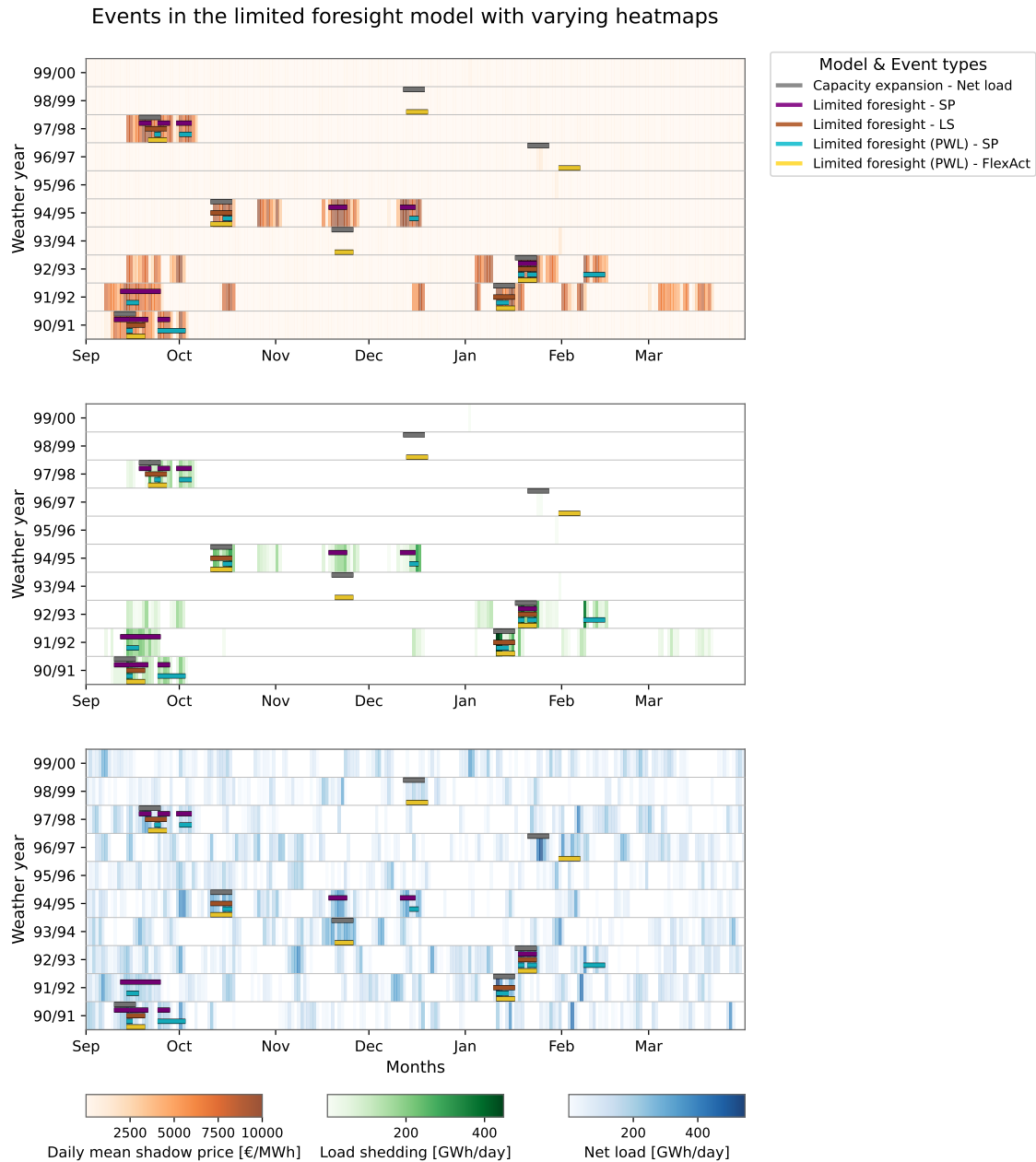


Figure 4.6: Identified events for the limited foresight models with heatmap backgrounds from load shedding, net load and shadow prices. Showing months from September to April.

Abbreviations explained: SP = Shadow price event, LS = Load shedding event, LF = Limited foresight model, PWL = Piece-wise linear model, FlexAct = Flexibility activation event.

Flexible demand activation

The period where most of the flexible demand is utilised is the so-called flex activation events. Generally, this period aligns well with the load shedding events and also often with the net load

events, as exemplified in Figure 4.6. Furthermore, the activation often identifies the same periods between the two dispatch models. The general picture is that the flexibility is used instead of load shedding, and that this often occurs in periods with high net load. A full figure containing the flexibility activation events alongside load shedding and net load events can be seen in Figure B.3.

4.2.3 Event duration and frequency vs cost threshold for the SP method

To show the influence of the threshold parameter β , I ran the models with varying thresholds from 10-40% in steps of 5%. The metrics used to quantify the impact are the average yearly number of events and the average duration (length) of the events. Although there may be variations within each year, these metrics still provide an insightful indication of how the threshold is affecting identification. The resulting figures can be seen in Figure 4.7.

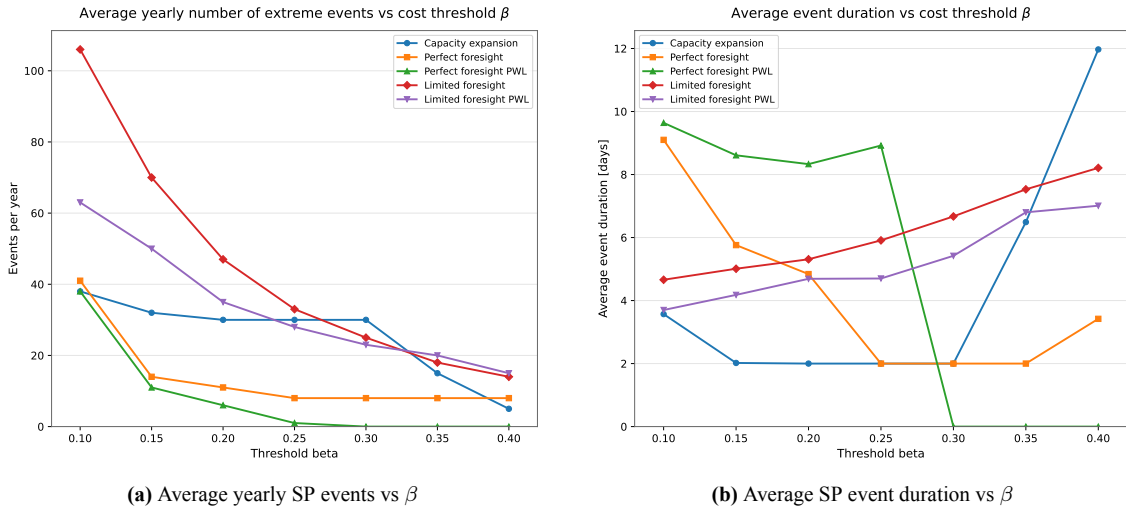


Figure 4.7: Average yearly number of shadow price events and average event duration across the models as a function of the threshold β . Average values taken across all 31 weather years

Figure 4.7a clearly shows a tendency for the number of events to diminish as the threshold is increased, particularly for the models with elastic demand. This is an expected result as the allowed length of each period T^{SP} is kept fixed; nonetheless, it is interesting to see that the CE model stays consistent until $\beta = 0.35$, opposed to limited foresight models, which have a more continuous decrease in events. The perfect foresight dispatch model is substantially below the other three models, with the PF-PWL model reaching 0 events with thresholds beyond 25%. The capacity expansion and dispatch models align quite well on the avg. events per year at $\beta = 0.20$, and looking at the duration plot in Figure 4.7b, we see that at this threshold the PF and LF models also align at around an average of 5 days per event.

Sensitivity from different capacities

In this section, the sensitivity of using different installed capacities in the dispatch models is shown. More specifically, I analyse the impacts on average event duration and average number of yearly events from using the shadow price method to identify events. For this analysis, the cost threshold is fixed at $\beta = 0.20$, and the used power capacities are the mean, 35th quantile and 75th quantile (see specific values in Table 3.2). The results are shown in Figure 4.8 and Figure 4.9.

The figures show that event identification is stable for most models, confirming the robustness of the shadow price-based method. The main change occurs when capacities increase: yearly average event detection for the LF models falls, and event durations shorten, which is also an expected outcome for installing more capacity.

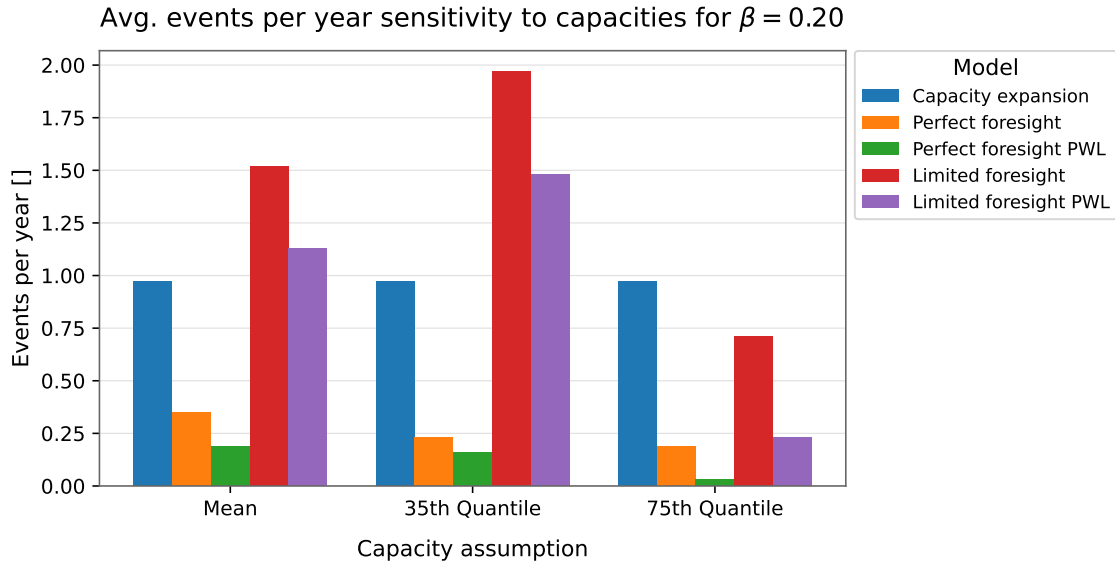


Figure 4.8: Sensitivity on average yearly number of events using the shadow prices method for different installed capacities in the dispatch models. Cost threshold used for the method: $\beta = 0.2$. The minimum and maximum event lengths used are 2-14 days.

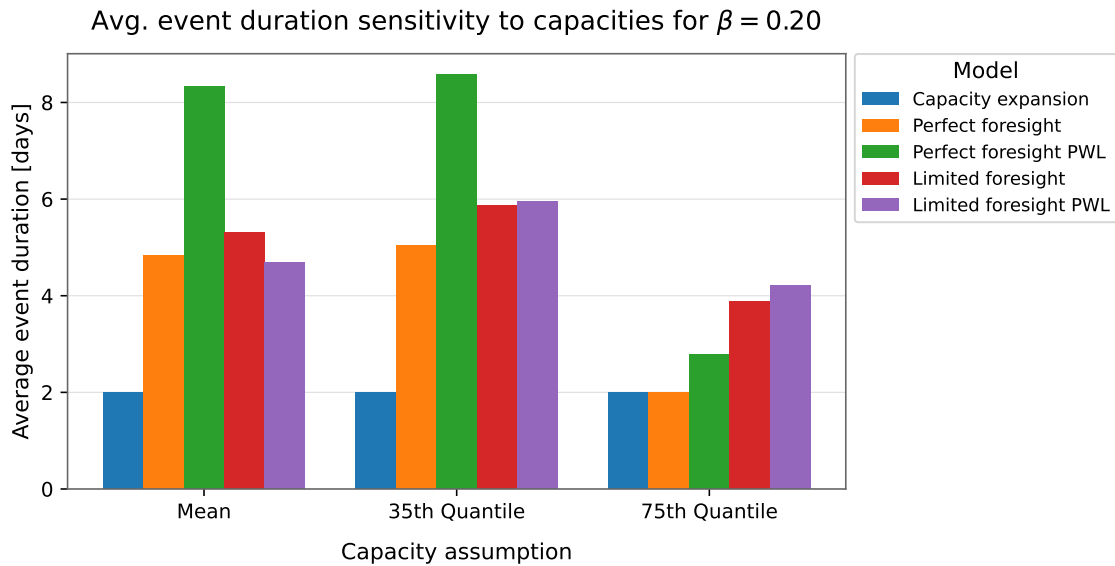


Figure 4.9: Sensitivity on average yearly event duration using the shadow prices method for different installed capacities in the dispatch models. Cost threshold $\beta = 0.2$. The minimum and maximum event lengths used are 2-14 days.

4.3 Extreme event alignment

4.3.1 Probability of events to overlap across models

In an attempt to analyse event alignment from a more statistical point of view, I created a matrix with probabilities for the events in the different model to overlap with each other, which is shown in Figure 4.10. The theory behind creating the matrix is briefly described in the following text.

The event overlap matrix is constructed as a row-conditional probability measure that quantifies the temporal alignment between the different types of extreme events identified in the five models. Each row corresponds to a specific model–event combination, while each column represents a reference event set. An entry in row i and column j expresses the probability that an event of type i overlaps in time with at least one event of type j .

Let $E_i(y)$ denote the set of events of type i occurring in year y , where each event $e \in E_i(y)$ is defined by a closed time interval $e = [t_e^{\text{start}}, t_e^{\text{end}}]$. Two events $e \in E_i(y)$ and $f \in E_j(y)$ are said to overlap if their time intervals intersect, i.e.

$$t_e^{\text{end}} \geq t_f^{\text{start}} \wedge t_f^{\text{end}} \geq t_e^{\text{start}}. \quad (4.1)$$

The probability of an overlap from event type i to event type j is defined as

$$P_{i \rightarrow j} = \frac{\sum_y |\{e \in E_i(y) \mid E_j(y)\}|}{\sum_y |E_i(y)|}. \quad (4.2)$$

Each event $e \in E_i(y)$ contributes at most once to the numerator, even if it overlaps multiple events in $E_j(y)$ such that the probability will always be a number between 0 and 1: $P_{i \rightarrow j} \in [0, 1]$. The resulting matrix is generally asymmetric, as the normalisation is performed with respect to the total number of events in the row category. Diagonal elements correspond to trivial self-overlaps, which are always equal to 1 and are therefore omitted from the visual representation for better clarity.

Interpretations from the overlap probability matrix

The insights to be gathered from the matrix are plentiful; however, I will try to highlight some of the key takeaways. The first aspect to consider is that the SP-CE events do not align very well with any of the other models (first row). All the white spaces mean probabilities below 0.25. This confirms the impression from Figure 4.5 where we saw how the events were both short and also present in years where the dispatch models did not detect any events. Being a simple yearly capacity expansion model having to recover investment cost during the year means that each model run have a substantial cost to recover compared to the dispatch models, where a year with no or little load shedding will mean more stable prices only coming from the variable costs, as also seen by the price duration curves in subsection 4.1.1. Nevertheless, we do see that the SP events from the PF model align very often with the capacity expansion model (third row); this contradiction in probability comes from the fact that the matrix is row conditional and that there are fewer PF-SP events compared to CE-SP events.

Another takeaway is that the PF-SP-PWL events in row 5 align completely with the PF-SP events and 83% with the PF flexible activation event (PF-FlexAct-PWL), suggesting that the flexible demand is being utilised in the extreme periods, causing higher prices. If we look at PF-FlexAct-PWL and LF-FlexAct-PWL events (row 5 and bottom row) we see that they both align 64% of the time with the CE net load events (2. column). This tells us that in over half of the years, the period where the flexible demand is most frequently utilised happens during the same period as the highest net load. This was also seen visibly in Figure 4.6 for the limited foresight (PWL) model with flexible demand.

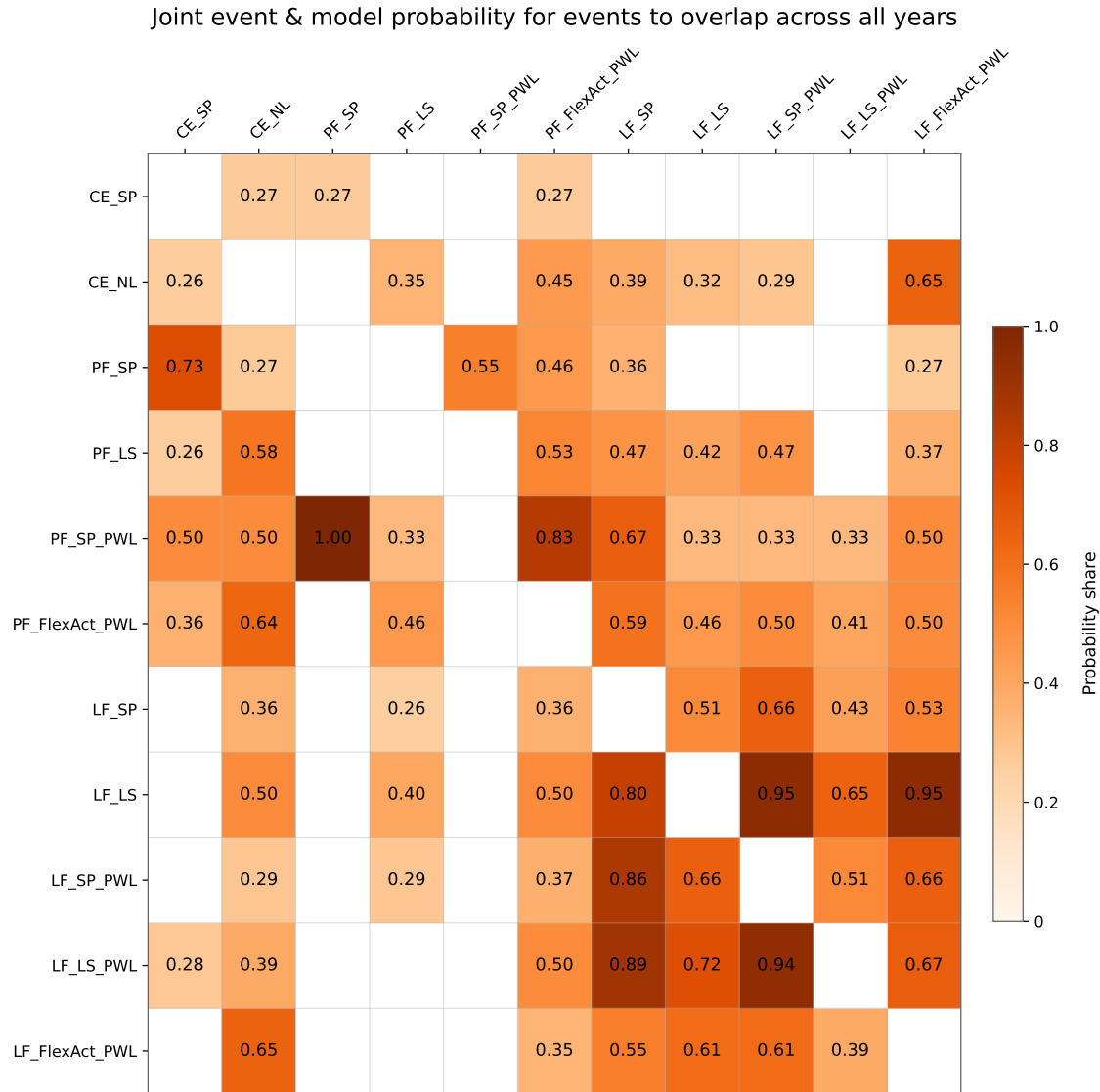


Figure 4.10: Probability heatmap matrix made from aggregated values over all simulated years. The matrix has been filtered to show only probabilities above 0.25. Additionally, the diagonal is removed as it is always 1 and thus trivial.

How to read the matrix: The matrix is *row-conditional*, meaning it should be read from the rows perspective. Example: at row PF_SP, we see the probability of a 'Perfect foresight - Shadow price event' to overlap with all the events in the columns, i.e. 0.73 chance of a PF_SP event to align with a CE_SP event (first column).

All abbreviations explained: *Models:* CE = capacity expansion, PF = perfect foresight, LF = limited foresight, PWL = Flexible demand model w. piece-wise linear demand curve. *Events types:* SP = Shadow price, NL = net load, LS = load shedding, FlexAct = demand flexibility activation event.

A final observation concerns the general coherence among the limited foresight models, reflected in the many dark red fields and high probabilities in the bottom five rows and last five columns. This was something that we also saw for a subset of years in Figure 4.6, and this confirms that these events tend to overlap internally relatively often for all the years. Especially the LF-LS load shedding events (4. row from the bottom) showing overlap probabilities of 0.65-0.95 for all other LF events. This suggests that considering load shedding alone in the LF model captures extreme periods reasonably well. It is important to note that the probabilities reported here indicate only the presence of an overlap and do not specify whether this overlap spans the entire period or only part of a day. To account for this, one must look specifically at the identified periods, as was done in section 4.2.

4.3.2 Fitting the threshold β to match average yearly events and duration

Looking back at Figure 4.7, we see that at $\beta = 35\%$ the LF and CE model tend to agree on average yearly events and also for the duration, at approximately 0.5 events pr. year and 7 days (respectively). While the PF model is consistently below the other models, it actually has a similar average events and duration values at $\beta=15\%$. Thus, these β values will be chosen to analyse how the events align when the average yearly events and duration length are almost the same across all models. These are plotted in Figure 4.11

The figure (4.11) does not tell a very interesting story when compared to the shadow price events with $\beta = 20\%$ from Figure 4.5. Besides slightly fewer events from all models but the perfect foresight model, we see that the alignment of the events is similar or slightly worse than before. Only the PF model agrees with new overlaps with the LF events in a few new years, like 90/91 and 89/90. The most interesting observation concerns the capacity expansion events, whose duration is extended slightly in some years, suggesting that this threshold value may be better suited for comparison with dispatch model events using $\beta = 0.20$.

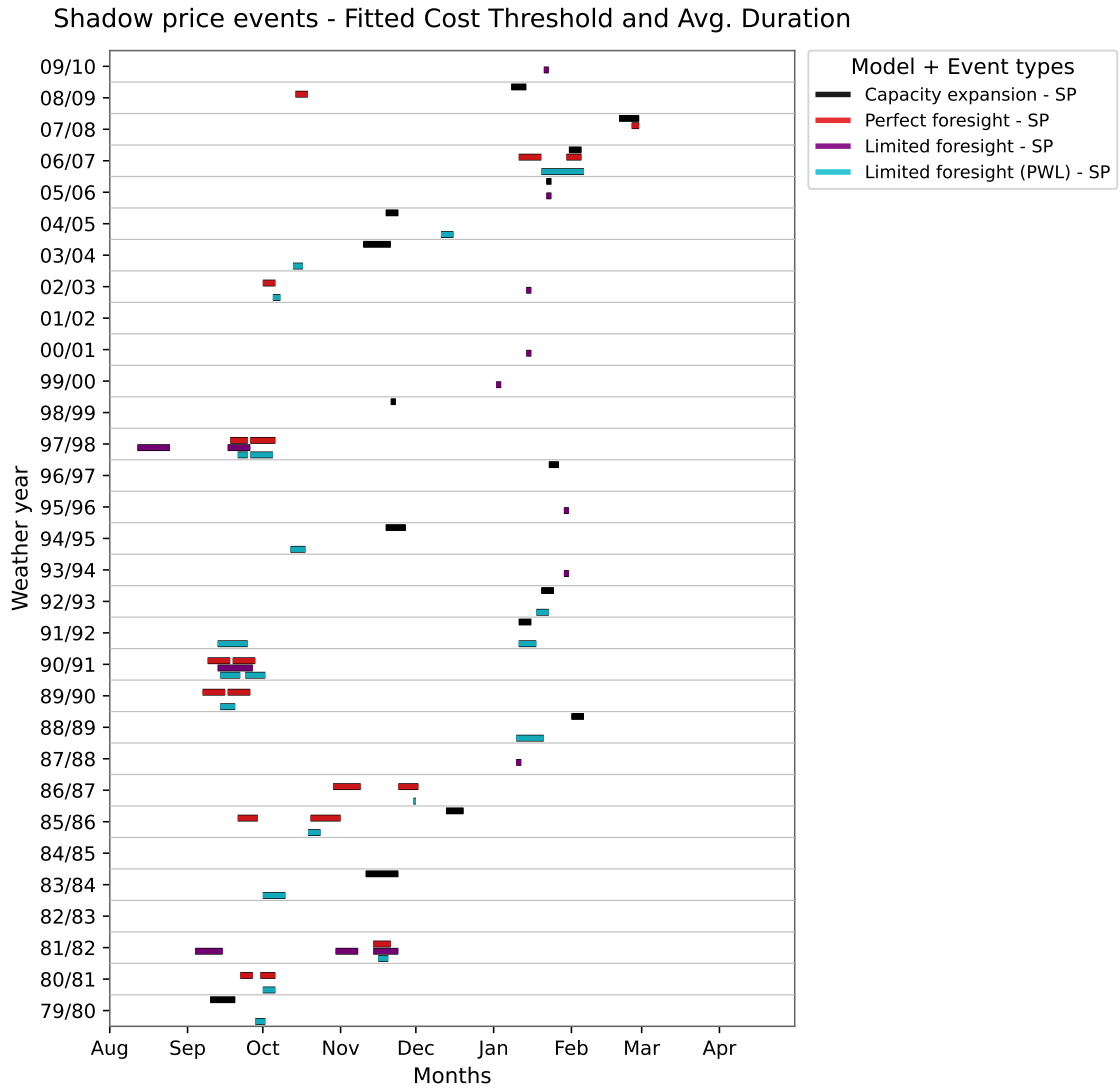


Figure 4.11: Shadow price events with fitted threshold values. $\beta = 35\%$ used for LF and CE models and $\beta = 15\%$ for PF models. **Abbreviations:** SP = Shadow price event, PWL = Piece-wise linear

4.4 System impacts of long-duration energy storage

4.4.1 Storage operation across the models

Figure 4.12 shows the yearly state of charge (SOC) profiles of long-duration storage for a subset of particularly challenging weather years. The figure displays the individual yearly SOC trajectories, together with the mean SOC across the selected years and the 25th and 75th percentiles to illustrate the spread in storage operation. The years included are those with ten or more identified extreme events across all event types, and the exact years are listed in the figure caption. The full figure containing all years can be viewed in the appendix (subsection B.2.1).

The figure provides an overview of the main storage operation patterns during these difficult years. Overall, the SOC profiles follow broadly similar seasonal patterns across all models, although notable differences appear in the details. For all models, some or most of the selected years experience periods of completely depleted storage ($SOC = 0$) during the winter half-year (September to March). From the lower quantile, it is evident that the perfect foresight model generally exhibits the lowest SOC levels compared to the capacity expansion model. This reflects the objective of the capacity expansion model, which minimises total system cost by installing just enough storage capacity to satisfy demand, resulting in SOC trajectories that just briefly reach zero, as seen by the grey lines.

The dispatch models, by contrast, operate with the mean capacity mix derived from the capacity expansion model and may therefore have sufficient storage capacity to avoid full depletion in some years. For the limited foresight models, longer and more frequent periods of empty storage are observed. One contributing factor is the lower initial state of charge, which directly reduces the amount of available energy during periods of low renewable generation. In fact, among the years shown, the perfect foresight models only rarely start with a lower initial SOC than the limited foresight model. In addition, the limited foresight models tend to deplete storage more rapidly, reflecting their inability to anticipate future scarcity and thus a greater propensity to use stored energy early. Finally, the LF models often end the year with a higher SOC, as they are not constrained by a cyclic end-of-year condition. This behaviour again reflects the lack of information about future system conditions, both scarce and abundant.

The final point to emphasise concerns the difference between models with flexible demand and those with inelastic demand. Based on this figure, no substantial difference in storage operation is observed between the two cases. The only notable deviation is found for the flexible perfect foresight model, where the SOC tends to be slightly lower than in the corresponding inelastic model. This behaviour is likely due to the model utilising flexible demand in segment 3 at prices below that of discharging the storage, which allows the model to keep the storage more depleted without reaching energy scarcity, and thereby reduce total system cost.

4.4.2 Storage operation vs extreme events

To illustrate the relationship between storage state of charge and the identified extreme events across the models, these quantities are shown jointly for two representative problematic years in Figure 4.13 and Figure 4.14. The figures allow for a clearer comparison of how SOC trajectories generally follow similar seasonal patterns across models, while still exhibiting important differences in the timing and identification of extreme events.

As discussed in section 4.2, there is a tendency for events identified under limited foresight to occur earlier in the year than those identified by the capacity expansion model. This behaviour is clearly illustrated for the 1980/1981 weather year in Figure 4.13. Here, the LF model identifies an event in early October, whereas the CE model identifies events in January and February. The LF event coincides with a period in which the storage is fully depleted, indicating a state of energy scarcity. Although the SOC in the CE and perfect foresight models is also low during this period, their use of

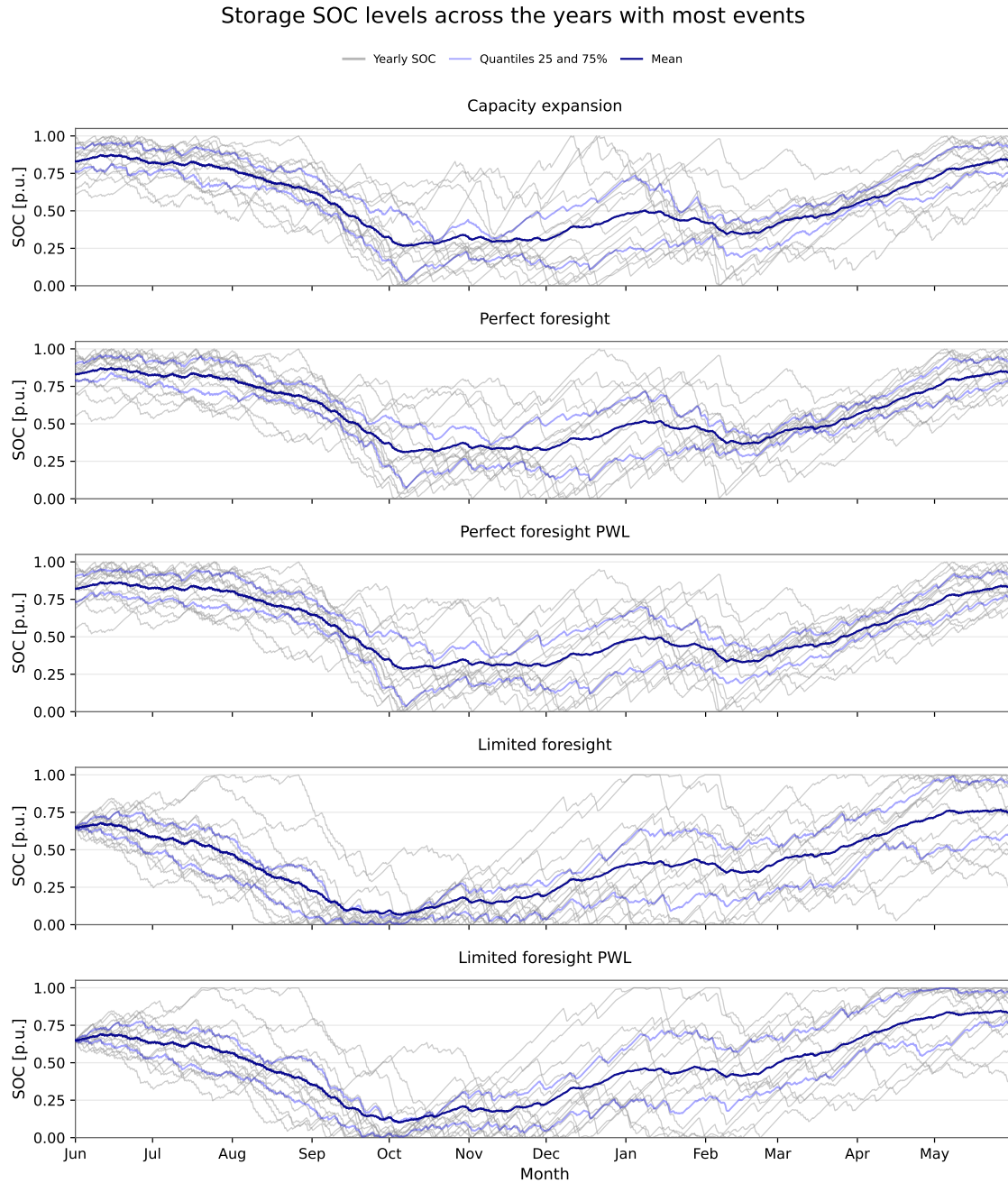


Figure 4.12: SOC yearly operation of the years with the most events identified across all event types. Showing the actual SOC profiles, the mean SOC across the selected years, and the 25th and 75th quantiles to enhance the variation of the profiles.

The plotted years are: 1979, 1980, 1981, 1983, 1985, 1986, 1988, 1989, 1990, 1992, 1994, 1997, 2002, 2003, 2006

perfect foresight allows them to plan sufficient stored energy to pass through the episode without fully depleting the storage, and the period is therefore not identified as an extreme event. It is worth noting that this is a very sensitive result, and that a small reduction in variable renewable energy production during this period would likely have resulted in an extreme shadow price event also for the CE and PF models. In addition, the CE model identifies a short shadow price event during a period when the storage is actively discharging but remains more than half full, at approximately 60% SOC. Given that these events coincide with very high prices, this suggests that the event occurs during a peak load situation characterised by low VRE production and maximum operation of discharging the storage. While such an event is system-defining from the perspective of the capacity expansion model, it may not represent a period of prolonged energy scarcity or sustained energy drought.

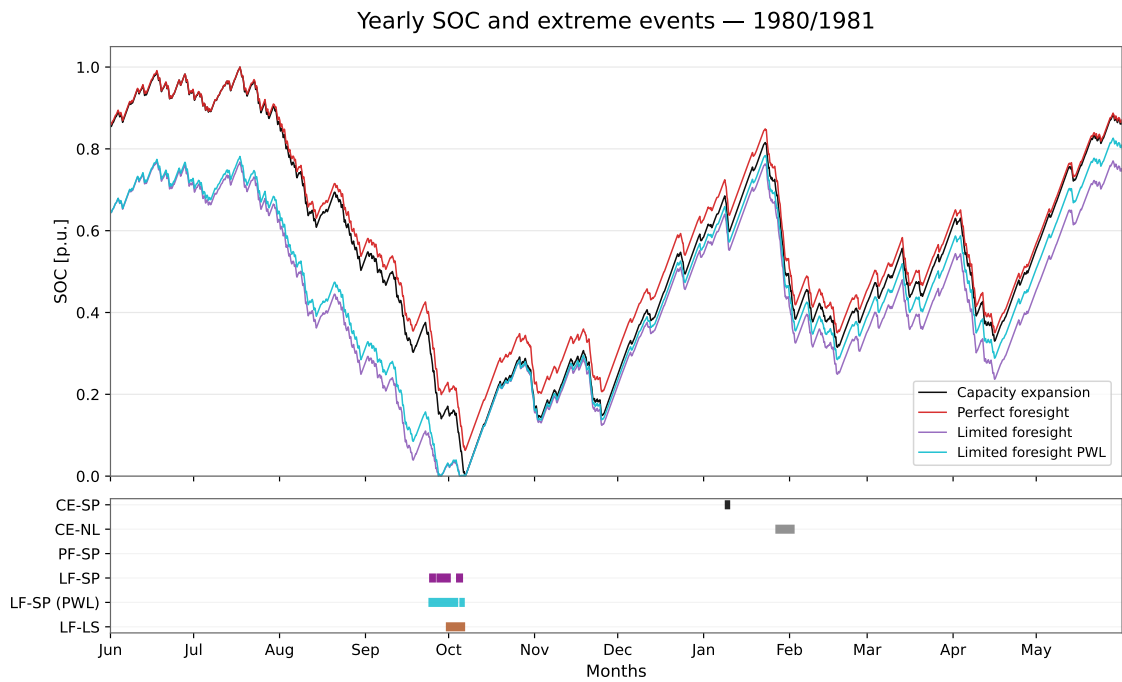


Figure 4.13: SOC operations across models combined with selected events for year 1980/1981.

The abbreviations yield: CE = Capacity expansion, PF = Perfect foresight, LF = Limited foresight, PWL = Piece-wise linear, SP = Shadow price events, NL = Net load event, LS = Load shedding event

In Figure 4.14, an example is shown where the LF and CE models identify shadow price events during the same period of low VRE production in September. Due to differences in the initial state of charge, the storage is fully depleted only in the LF model, whereas the CE model identifies the period as critical based on generation based energy scarcity. This interpretation is supported by the net load event, which shows a pronounced peak during the same period. In addition, the LF model identifies a second period of energy scarcity later in the year. As with the event shown in Figure 4.13, this December period corresponds to a short interval during which the storage becomes depleted across all models. Nonetheless, the CE and PF models again appear to rely on perfect foresight to maintain just enough stored energy to pass through this episode until the model can recharge the storage later in the year.

Several of the other problematic years follow similar trends, although in some years the events align across all models, only to be identified slightly earlier by the limited foresight model. An example can be seen in the appendix in Figure B.8 for 2006/2007.

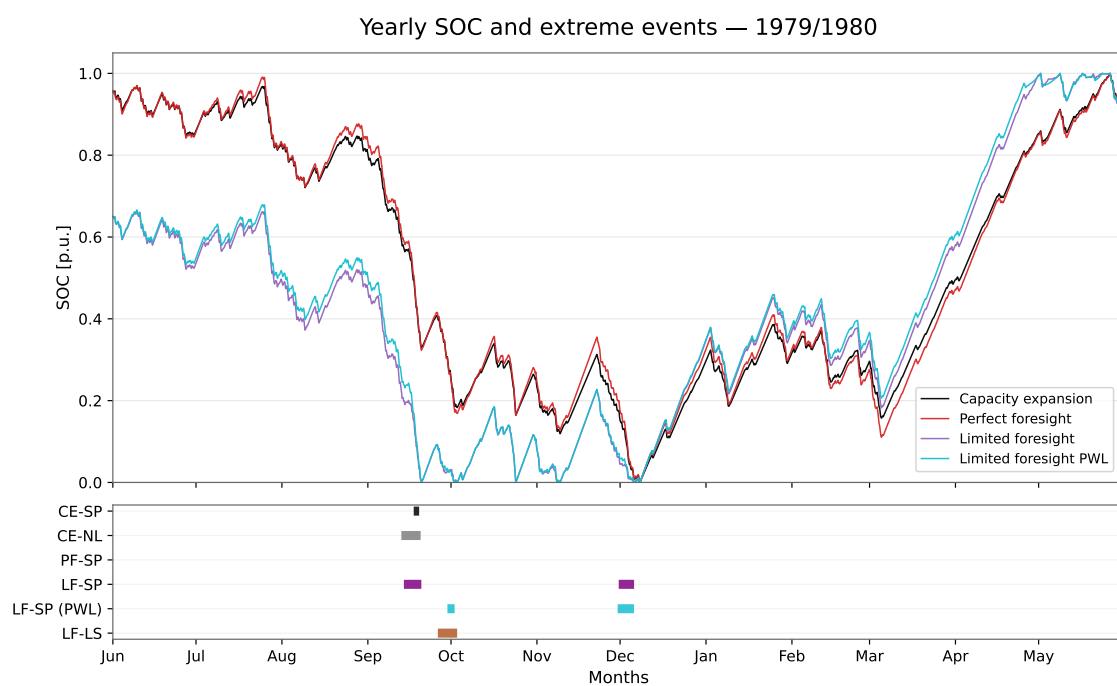


Figure 4.14: SOC operations across models combined with selected events for year 1979/1980.

The abbreviations yield: CE = Capacity expansion, PF = Perfect foresight, LF = Limited foresight, PWL = Piece-wise linear, SP = Shadow price events, NL = Net load event, LS = Load shedding event

5 Discussion

5.1 Interpretation of the key results

One of the objectives of this thesis (1.4) is to analyse whether the shadow price based extreme event identification method proposed by Grochowicz, Greevenbroek, and Bloomfield [9] can identify events in economic dispatch models, which is confirmed by the results of the thesis. The results show that the method identifies periods of energy scarcity (4.4.2), and that these periods tend to align with periods of high net load and load shedding (4.5), as also suggested in [9], where the method was applied to a capacity expansion model.

The shadow price method is less effective for the perfect foresight model, in which storage tends to set sustained high prices as opportunity costs, thereby obscuring the interpretation of high prices as indicators of system stress. Despite this, the events identified in the perfect foresight model align significantly with those identified by the capacity expansion model. One of the key results when applying the method to models with limited foresight is that it identifies events earlier than the capacity expansion model and, in some years, identifies a larger number of events.

A sensitivity analysis of the implications of the method's total cost threshold on average event duration and number of events shows that this parameter is highly influential. It therefore needs to be chosen carefully depending on the objective of the analysis, that is, whether the aim is to identify only the most extreme and severe periods or to identify multiple candidates for extreme events. In a broader energy system planning context, a lower threshold can be used to identify different potential triggers of such events, whether these arise from transmission bottlenecks, insufficient installed capacity, or energy storage depletion. Using a higher threshold helps to identify longer and more severe events, such as those described by [10], which stem from prolonged compounding weather conditions leading to sustained energy droughts. Defining the cost threshold as a percentage of the total system cost in each year proves to be an effective way of applying the method to systems with more arbitrary definitions and is also useful for comparing event magnitudes between models that do not share the same total cost scale, such as the capacity expansion and dispatch models used in this study.

Comparing the identification methods against each other shows that there are generally many overlapping events, particularly load shedding events, in the limited foresight model. This indicates that load shedding performs well as an event identification method in the limited foresight model (4.10) and also aligns well with net load events. In addition, net load events are generally effective at capturing the most extreme events in the limited foresight model, but they also exhibit the greatest variability in identification. This is likely an artefact of the event definition used in this thesis, which enforces the identification of one net load event per year with a fixed duration. Nevertheless, the actual net load varies significantly between years. The net load method lacks information about storage availability and its ability to contribute during periods of high net load. Extending the net load metric to include information about storage dispatch and or state of energy could allow it to identify events that more closely align with those identified through high prices and load shedding.

Flexible demand is successfully modelled through the introduction of piece-wise linear demand price elasticity, implemented via a load shedding generator as described in [19], representing different consumer segments. Introducing flexible demand reduces both the amount of load shedding and the number of events identified by the shadow price method. In addition, the events identified in these models largely coincide with the load shedding and shadow price events identified in models with inelastic demand. Flexible demand also has a positive effect on price formation (4.1.1), although large price steps still remain a prominent feature of the models. Interpreting de-

mand elasticity as a means of reducing identified extreme events is an important result for future energy system planning, as it may influence capacity planning decisions aimed at avoiding such events. Further analysis of the cost of utilising flexible demand as an alternative to full capacity expansion under completely inelastic demand may therefore prove to reduce total system costs.

The results show the importance of long-duration energy storage in terms of both extreme event identification and price formation. As also discussed in recent literature [15, 19], the bidding behaviour of long-duration energy storage is highly influential in systems with high variable renewable energy penetration and low marginal electricity production costs. The price formation results in the dispatch models in this study show the same tendency. Furthermore, using a mean marginal storage value as the marginal cost for long-duration energy storage proves sufficient to operate the storage in a similar manner in models with limited foresight.

The importance of the storage level is further reinforced by comparing how shadow price and load shedding events are identified in limited foresight models during periods with low storage energy levels. In contrast, the net load events and shadow price events identified in the capacity expansion model show that events are not always directly tied to the storage energy level, but also to the available dispatch power of the storage, represented by the installed fuel cell capacity. Suggesting that events in these models are more likely to represent power capacity inadequacy opposed to energy capacity. Moreover, the limited foresight assumption, which removes the ability to operate storage cyclically (enforcing equal initial and final SOC), emerges as an important trigger of events in the model. This is also likely to contribute to the earlier identification of events in the limited foresight models, highlighting the initial state of charge as a key parameter for both the analysis and event identification.

From a broader energy system modelling perspective, these results illustrate how the assumption of perfect foresight may fail to clearly identify sensitive periods that would likely arise under more realistic operational conditions without full foresight. This effect is particularly visible in fig. 4.13 and 4.12, where the storage level in the capacity expansion and perfect foresight dispatch models approaches zero multiple times without being categorised as an event, as the system retains just enough energy to remain feasible. This highlights the sensitivity of numerical solvers that optimise energy systems solely under a cost minimisation objective. Designing a system that operates close to storage depletion is highly sensitive to uncertainties arising from imperfect knowledge of future weather conditions. Employing a limited foresight modelling approach is one possible way to account for this, while alternative approaches include robust optimisation, stochastic programming, or solving for near-optimal solution spaces [40, 24].

5.2 Important limitations of this study

In this thesis, several simplifications are applied to focus the analysis and clarify the effects of changing the modelling assumptions related to limited foresight and demand flexibility. Nevertheless, some limitations remain important to highlight in light of the results, which are briefly discussed in this section.

All models employed in this thesis are stylised and highly simplified, encompassing only a limited set of technologies and excluding any representation of the transmission network. The absence of transmission lines, the absence of additional technologies, and the lack of spatial differentiation in demand and production are among the most significant limitations of this work. Consequently, the results presented here do not guarantee that applying the same extreme event identification methodologies, together with assumptions of limited foresight and demand elasticity, to a larger and more detailed model of a real energy system would yield comparable outcomes.

Nonetheless, the findings indicate that the proposed methodology has potential practical relevance for real-world energy system planning. In particular, it can be used to generate insights into the

design of robust energy systems capable of withstanding extreme events, which represent a substantial risk in power systems with a high share of renewable energy sources.

Furthermore, the installed capacities determined by the capacity expansion model should not be interpreted as realistic. They ignore existing generation capacities, such as nuclear and hydropower, as well as spatial and land use constraints, which are important factors in assessing whether such levels of variable renewable energy production and storage can be realistically deployed. In addition, the system cost represents an abstract and arbitrary value and cannot be directly compared to real system costs. The results, therefore, illustrate modelling dynamics rather than providing prescriptive guidance on how the real energy system of Spain should be designed.

Another limitation concerns the data used in the analysis. Ideally, the system's performance is tested over a larger set of weather years. Performing an in-sample and out-of-sample analysis across more weather years would ensure that the models are exposed to weather data that is not used in the optimisation and configuration of the system. Although using mean installed capacities represents an attempt to test the system under unfamiliar conditions, the robustness of the results would improve by evaluating the models using a broader range of weather data. The use of demand data from only a single year is also a limitation, as variations in demand patterns may affect how extreme events are identified.

In addition, restricting the temporal scope of the simulations to single-year periods limits the ability to identify interannual weather trends, which may lead to compounding conditions and more severe extreme events. Starting the simulations during summer, when weather conditions are typically more stable, mitigates some interannual effects, but this remains an important limitation when interpreting the results.

Regarding the modelling of flexible demand, a key limitation is that the analysis does not explicitly represent when the deferred demand is ultimately satisfied. In practice, flexible demand is more likely to shift to lower-price periods than to be removed entirely. Excluding this effect may bias the results, as higher demand later can limit storage recharging and change the frequency of extreme periods. The current modelling assumes that once demand is curtailed at a high price, it is fully met and then leaves the system, whereas in reality, flexible demand is typically shifted over time rather than permanently reduced. One approach to represent this behaviour is the use of cross elasticity, as also demonstrated in [19], which modifies demand elasticity in subsequent hours following a reduction in demand. This approach is shown to affect peak prices and reduce the total utilisation of flexible demand, which is particularly relevant given the influence of high prices on the shadow price-based event identification method.

Furthermore, the analysis assumes that demand flexibility is uniform across the year and throughout the day. The modelled demand price elasticities are constant over time, which is unlikely to hold in real systems. In addition, both the elasticities of the demand segments and the energy volumes they represent are subject to considerable uncertainty. As discussed in the implementation chapter, such parameters are difficult to quantify. Consequently, the piece-wise linear demand curve applied in this thesis is unlikely to fully represent the true shape of real electricity demand.

Lastly, several limitations related to the event identification methods are highlighted. The methods used to identify net load, load shedding, and flexible demand events are comparatively simplified relative to the shadow price-based identification method. In particular, the assumption of a fixed event duration plays an important role in defining the identified events. As shown in Figure 4.6, which illustrates events in the limited foresight model, the heatmaps for load shedding and net load indicate that both shorter and longer events may be identified if event durations are allowed to vary.

In addition, these methods are restricted to identifying a single event per year. This limits direct comparability with the shadow price method, which is capable of identifying multiple events within

a year and does so in several cases. Allowing multiple events for the alternative identification methods would provide a more comprehensive comparison. Furthermore, the thresholds applied to these methods are set relatively low, and the net load method does not include an explicit threshold, resulting in the identification of one event per year. Introducing stricter thresholds would likely reduce the number of identified events and isolate only the most severe system events. To focus the analysis more explicitly on extreme events, better tuned thresholds should therefore be applied in future work. Nevertheless, the chosen approach provides a useful basis for comparing the timing of periods in which these metrics reach their highest values.

Finally, the event identification does not include a detailed analysis of the extent to which events overlap in time, such as whether overlaps occur over hours, days, or entire periods. The probability matrix presented in Figure 4.10 does not capture this temporal dimension. However, from an energy system planning perspective, a more detailed analysis would likely be conducted following the identification of periods with multiple coinciding events, even if their temporal overlap is limited. The current approach therefore remains useful for capturing the broader picture of extreme and potentially system defining events.

5.3 Future work and outlook

A natural next step to extend the findings of this thesis is to assess the results robustness under increasing system complexity. One extension would be to include additional power generation technologies, as well as other types of short and long-duration energy storage, such as battery systems and large-scale hydroelectric storage. Another step would be to increase the spatial resolution of the model by introducing multiple nodes and investigating how transmission constraints influence extreme event identification. Further extensions could broaden the system boundaries beyond electricity to include other sectors, such as heating. Moreover, introducing ramping constraints on generation technologies would allow an assessment of their impact on system operation, particularly under limited foresight.

Together, these extensions could provide further insight into which elements of an energy system most frequently drive extreme system events. A more detailed categorisation of event types could also be explored to assess which types of events different identification methods tend to capture. Such categories could, for example, include transmission-constrained events, storage-depletion events, high-demand events, low-generation-capacity events, high-net-load events, and compound events in which several of these conditions occur simultaneously.

Another promising direction is to further analyse extreme event identification using the shadow price-based method. More specifically, the approach could be tested on historical electricity price data to assess whether it aligns with commonly recognised 'dunkelflaute' periods. This would provide insight into whether the method is applicable not only to energy system models but also to real-world power systems.

More extensive analysis of long-duration energy storage could also be conducted. For instance, the limited foresight models could be run with varying initial states of charge to assess the influence of this parameter on system operation and event identification. In addition, using a more detailed marginal storage value time series as the marginal cost for long-duration energy storage in the limited foresight model could improve the representation of storage operation under partial knowledge of future conditions. One example would be to vary the marginal storage value seasonally to ensure that storage is better prepared for potential scarcity periods during winter.

The last recommendation is to extend the analysis to simulate multiple consecutive years which would enable the study of overlapping and compounding extreme events and provide further insight into how the timing and occurrence of such events differ between limited foresight and perfect foresight operations.

6 Conclusion

In this thesis, I investigate how extreme event identification methods are influenced by common modelling assumptions, specifically perfect foresight and inelastic demand. The analysis focuses on how these assumptions shape the identification of extreme system events and how they affect the perceived role of long-duration energy storage (LDES) under different modelling approaches. By comparing perfect foresight and limited foresight formulations, as well as inelastic and elastic demand representations, the thesis assesses how modelling choices influence both the detection and interpretation of extreme events in highly renewable energy systems.

The thesis shows that the extreme event identification method based on shadow prices is applicable to dispatch models. Across the analysed models, the identified events align well with other indicators of system stress, including periods of high net load, load shedding, and peak flexible demand activation. Comparing the method across the different models further shows that it performs more consistently under limited foresight, where identified events are clearer and more directly linked to resource scarcity conditions.

In contrast, the method proves more difficult to interpret in the perfect foresight dispatch model. Under perfect foresight, storage operation leads to sustained high prices driven by opportunity costs rather than acute scarcity, which complicates the identification of extreme events. Load shedding is also more evenly distributed over the year and does not necessarily coincide with periods of high system stress, resulting in significantly fewer identified events. These findings indicate that perfect foresight can distort extreme event identification by smoothing scarcity over time, thereby obscuring the actually observed short-term stress periods.

The analysis of LDES highlights its central role in extreme event identification. In the limited foresight model, storage depletion is particularly important in defining extreme events, whereas in models with perfect foresight, events tend to be associated with the storage reaching its maximum discharge capacity rather than full depletion. In several years, storage operation is even optimised to approach depletion for a very brief period without triggering extreme events. Given the stochastic nature of weather and, thereby, renewable energy generation, this behaviour suggests that perfect foresight may produce misleading signals regarding the timing and severity of extreme events, with implications for capacity planning decisions aimed at avoiding such scarcity periods.

The results for the flexible demand models show that introducing demand elasticity reduces the number and severity of identified extreme events. Periods with the highest flexible demand utilisation tend to align with other event types, including peak net load and load shedding events, proving that demand flexibility absorbs part of the system stress that would otherwise be reflected as extreme events. The elasticity effects on price smoothing also reduce the shadow price-induced events. This suggests that modelling demand as partly elastic can alter both the occurrence and interpretation of extreme events, with potential consequences for energy planning conclusions.

Taken together, the findings of this thesis show that assumptions regarding foresight and demand elasticity have a strong influence on extreme system event identification in highly renewable energy systems, and that the optimisation and operation of long-duration energy storage plays a crucial role in determining when and how such events are detected. By highlighting how extreme event metrics respond to different modelling choices, this thesis contributes to a more transparent interpretation of system stress indicators. Future work could extend the analysis by increasing model complexity, while incorporating limited foresight and elastic demand in planning models may support more robust and weather-resilient energy system designs under high shares of variable renewable energy.

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A Modelling setup and implementation

A.1 Capacity expansion model

A.1.1 Optimised capacities per year

Table A.1: Optimal yearly capacities for capacity expansion model

Year	onwind	solar	electrolysis	fuel cell	Hydrogen storage
1979	142,518	29,015	11,771	50,254	10,936,070
1980	144,201	25,796	12,297	54,486	9,505,567
1981	143,181	27,365	11,598	54,093	12,604,000
1982	147,516	26,286	13,710	56,181	5,130,176
1983	152,371	24,665	11,143	51,457	10,466,737
1984	140,656	24,769	11,368	51,600	9,115,841
1985	145,546	26,277	10,867	54,081	10,769,544
1986	146,206	24,066	11,165	55,229	11,703,402
1987	147,707	16,281	12,004	57,447	6,093,542
1988	156,364	30,724	12,961	58,090	9,613,210
1989	148,843	24,456	11,911	52,470	10,563,544
1990	143,509	24,814	11,460	53,777	12,201,175
1991	148,620	27,285	13,201	62,172	9,534,091
1992	152,267	30,625	13,255	54,384	6,414,591
1993	132,895	27,483	13,045	64,579	9,337,394
1994	139,718	31,151	11,923	58,095	15,637,770
1995	141,274	23,815	10,848	57,147	7,788,086
1996	141,171	29,965	13,241	59,190	7,615,737
1997	141,284	24,939	10,518	52,898	12,754,582
1998	136,857	20,072	10,276	53,578	3,290,148
1999	140,572	26,011	12,920	57,864	7,512,154
2000	129,483	20,766	9,722	64,145	12,756,271
2001	143,398	24,571	12,620	49,885	4,933,746
2002	134,908	27,927	11,924	65,580	11,900,895
2003	148,584	24,424	11,440	52,618	12,195,872
2004	141,858	27,894	13,130	54,083	8,785,596
2005	144,414	24,274	10,970	56,367	9,542,112
2006	145,839	28,736	13,972	61,026	7,500,371
2007	141,197	31,060	12,693	61,113	9,283,882
2008	147,041	19,763	10,830	52,035	8,347,617
2009	141,437	19,581	10,484	58,071	9,849,561

The optimised capacities over the 32 weather years

A.1.2 Mean MSV, mean shadow price and objective values in the capacity expansion problem

Table A.2: Summary of key results from the capacity expansion model across weather years.

<i>Unit</i>	Mean MSV [EUR/MWh]	Mean Shadow Price [EUR/MWh]	Objective [MEUR]
79/80	86.46	100.31	25817.23
80/81	83.04	97.04	26189.19
81/82	85.75	101.06	26447.49
82/83	86.22	98.37	26355.10
83/84	88.17	102.65	26549.72
84/85	84.79	98.33	25220.90
85/86	86.58	100.35	26219.96
86/87	87.01	101.35	26502.24
87/88	83.33	95.05	25841.94
88/89	89.22	102.83	28213.30
89/90	86.15	100.89	26455.27
90/91	85.17	99.45	26235.09
91/92	87.32	101.14	27749.39
92/93	86.48	98.35	26951.98
93/94	86.05	99.12	26372.66
94/95	87.55	103.59	27242.87
95/96	84.42	96.58	25556.39
96/97	88.20	98.57	26512.46
97/98	83.26	98.47	25818.63
98/99	81.15	89.61	23732.17
99/00	85.87	99.67	26028.78
00/01	81.59	96.05	25519.64
01/02	83.90	94.29	24903.24
02/03	86.30	100.87	26882.42
03/04	87.42	101.80	26596.96
04/05	85.76	97.42	26060.78
05/06	81.00	93.23	26096.09
06/07	87.51	99.27	27253.65
07/08	88.82	100.86	26927.34
08/09	82.21	93.94	25451.65
09/10	83.69	96.43	25710.17

A.1.3 Economic dispatch problem formulation with storage

$$\min_{g_{i,t}, s_t^{\text{ch}}, s_t^{\text{dis}}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} c_i^{\text{var}} g_{i,t} \quad (\text{A.1})$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}} g_{i,t} + s_t^{\text{dis}} - s_t^{\text{ch}} = D_t, \quad \forall t \in \mathcal{T} \quad (\text{A.2})$$

$$0 \leq g_{i,t} \leq G_i, \quad \forall i, t \in \mathcal{I}, \mathcal{T} \quad (\text{A.3})$$

$$E_t = E_{t-1} + \eta^{\text{ch}} s_t^{\text{ch}} - \frac{1}{\eta^{\text{dis}}} s_t^{\text{dis}}, \quad \forall t \in \mathcal{T} \quad (\text{A.4})$$

$$0 \leq s_t^{\text{ch}} \leq S^{\text{ch}}, \quad \forall t \in \mathcal{T} \quad (\text{A.5})$$

$$0 \leq s_t^{\text{dis}} \leq S^{\text{dis}}, \quad \forall t \in \mathcal{T} \quad (\text{A.6})$$

$$0 \leq E_t \leq E^{\text{max}}, \quad \forall t \in \mathcal{T} \quad (\text{A.7})$$

$$G, S, E, D, g, s, c \in \mathbb{R}^+ \quad (\text{A.8})$$

B Results

B.1 Event identification plots

B.1.1 Shadow price event with LF load shedding heatmap shown for the whole year

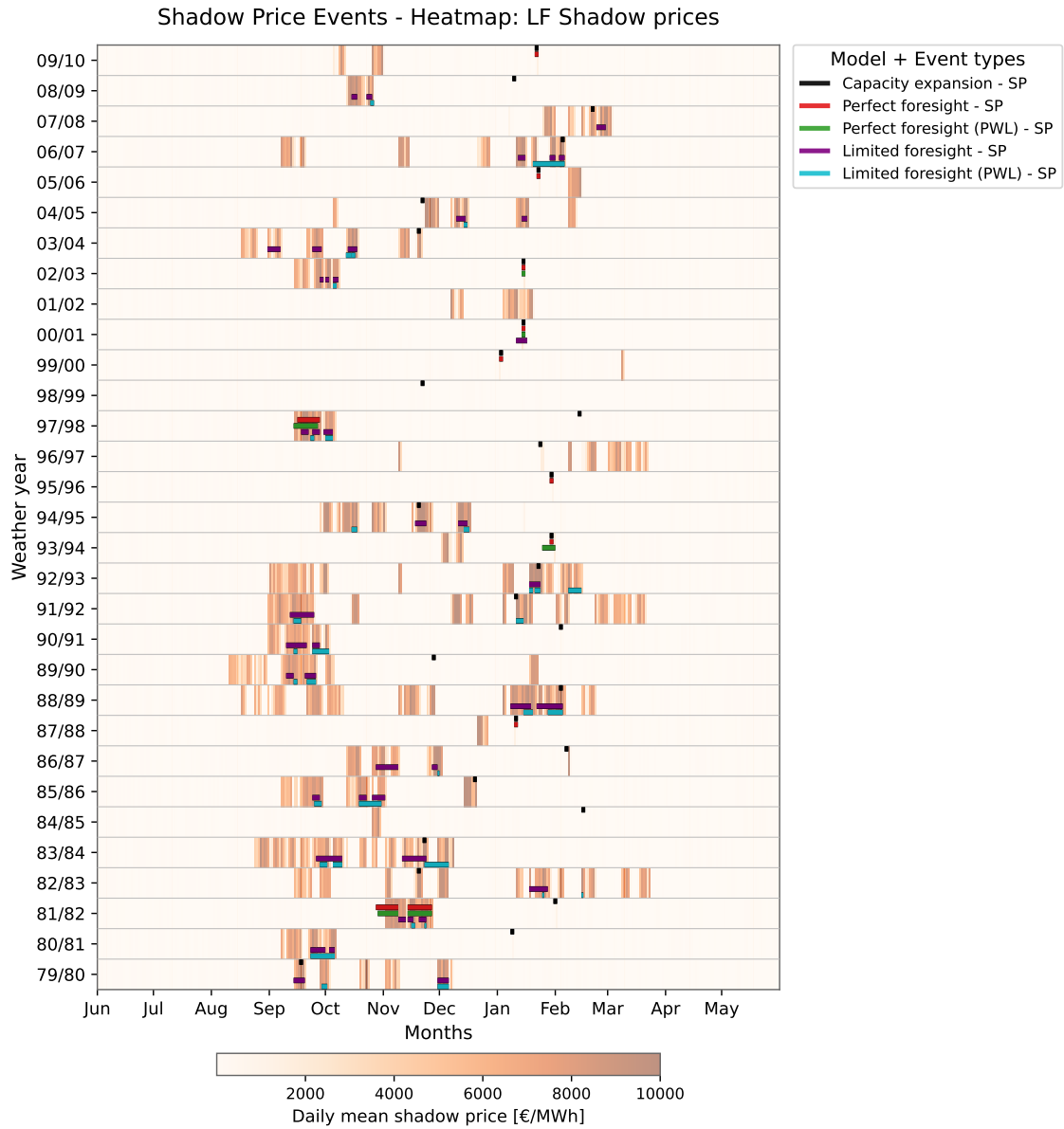


Figure B.1: Identified shadow price events with heatmap background from limited foresight model - All months

B.1.2 Shadow price events with heatmap from the shadow prices of the perfect foresight model

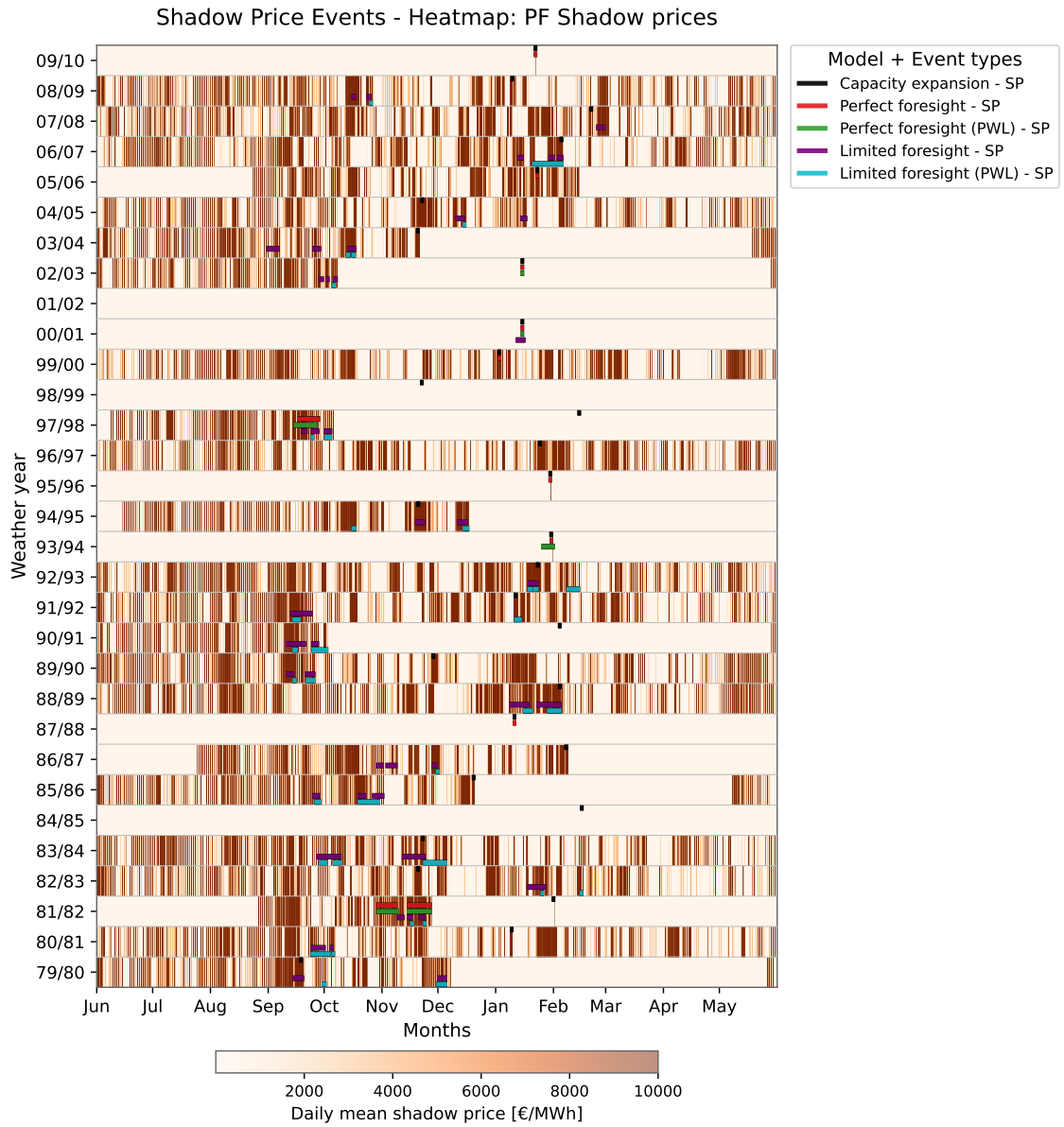


Figure B.2: Identified shadow price events with heatmap background from perfect foresight model - September to April

B.1.3 Flexible demand activation and load shedding events

Load shedding and Flex Activation events in LF model - Heatmap: LF load shedding

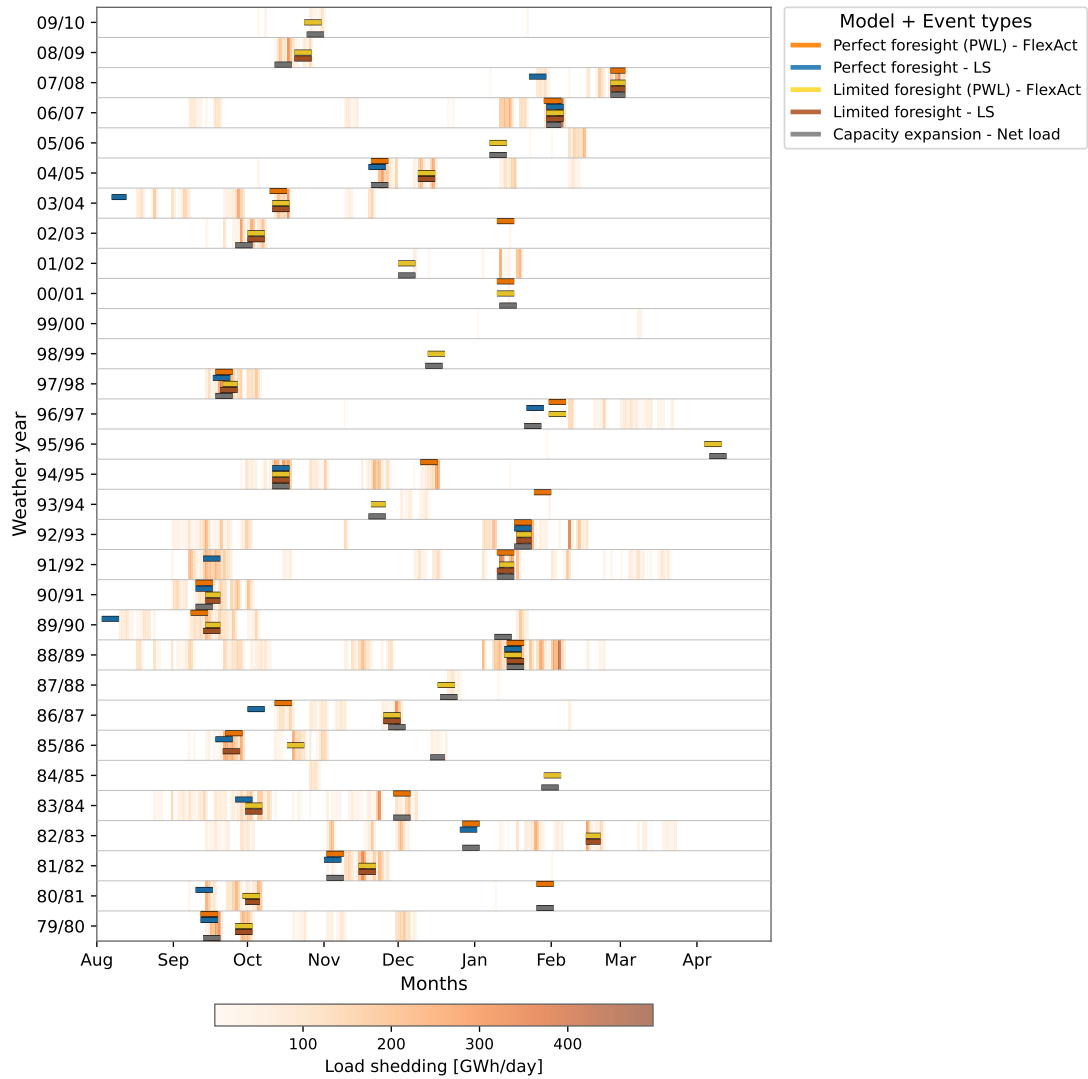


Figure B.3: Identified load shedding vs flexibility activation events in the dispatch models. With heatmap background from load shedding in the LF model - Showing months from August to May

B.1.4 Limited foresight model events with different heatmaps - whole year

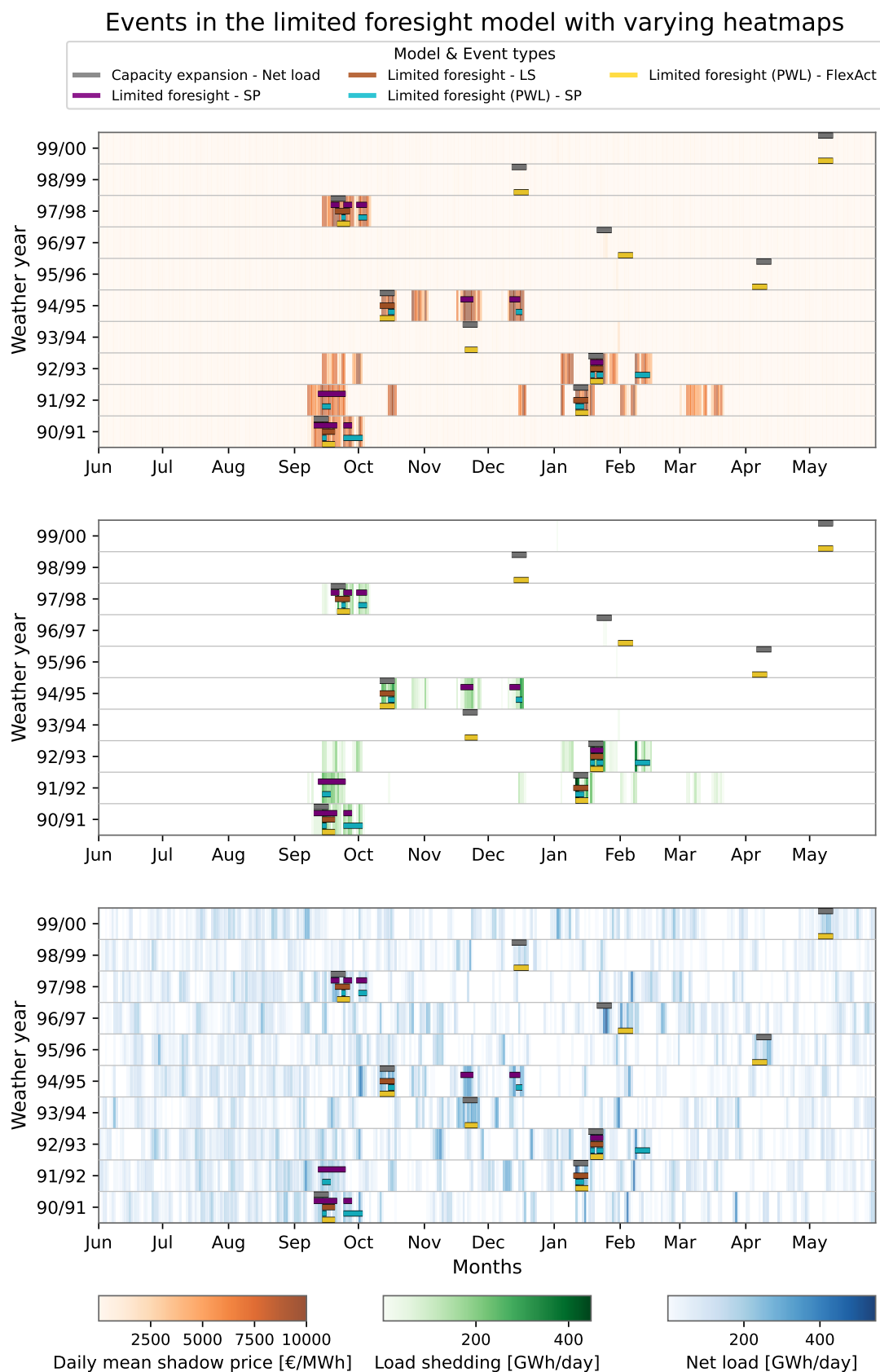


Figure B.4: Identified events for limited foresight model. With heatmap background from load shedding, net load and shadow prices. Showing full year

Abbreviations explained: SP = Shadow price event, LS = Load shedding event, PWL = Piece-wise linear model, FlexAct = Flexibility activation event

B.1.5 Limited foresight model events with different heatmaps - plots for the rest of the years

Events in the limited foresight model with varying heatmaps

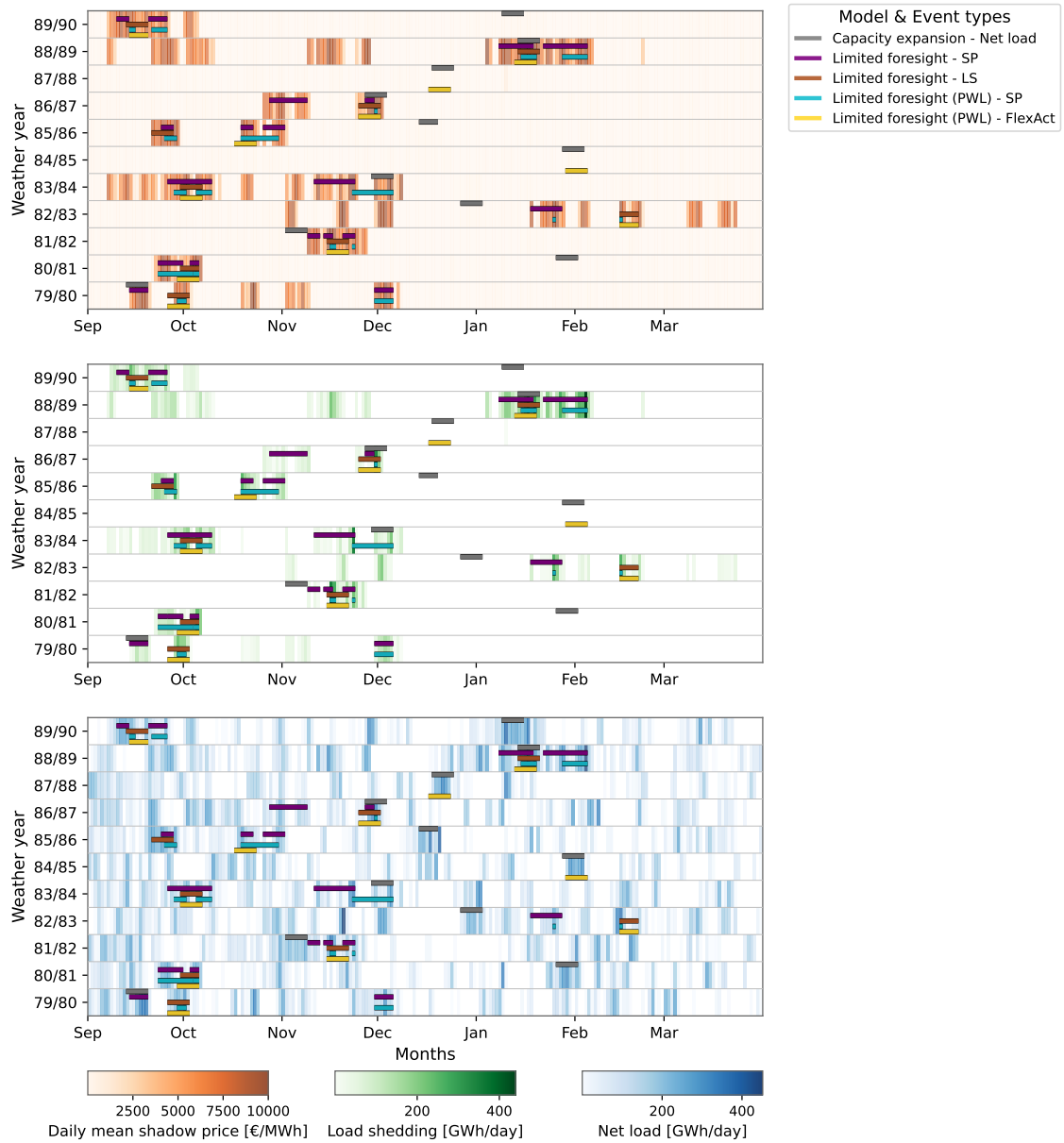


Figure B.5: Identified events for limited foresight models with heatmap backgrounds from load shedding, net load and shadow prices. Showing months from September to April.

Abbreviations explained: SP = Shadow price event, LS = Load shedding event, PWL = Piece-wise linear model, FlexAct = Flexibility activation event

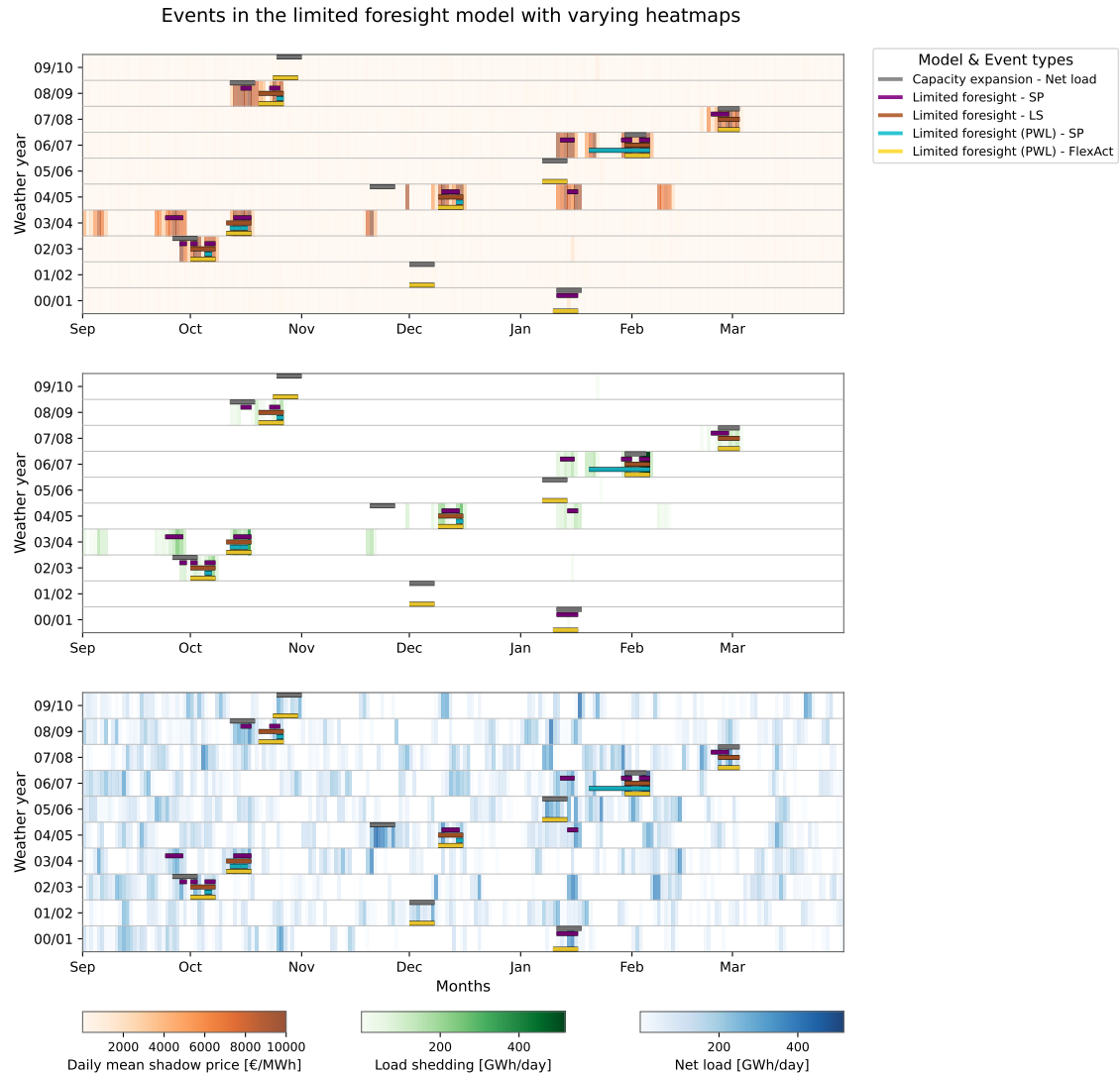


Figure B.6: Identified events for limited foresight models with heatmap backgrounds from load shedding, net load and shadow prices. Showing months from September to April.

Abbreviations explained: SP = Shadow price event, LS = Load shedding event, PWL = Piece-wise linear model, FlexAct = Flexibility activation event

B.2 SOC operation plots

B.2.1 Storage operation across the models for all weather years

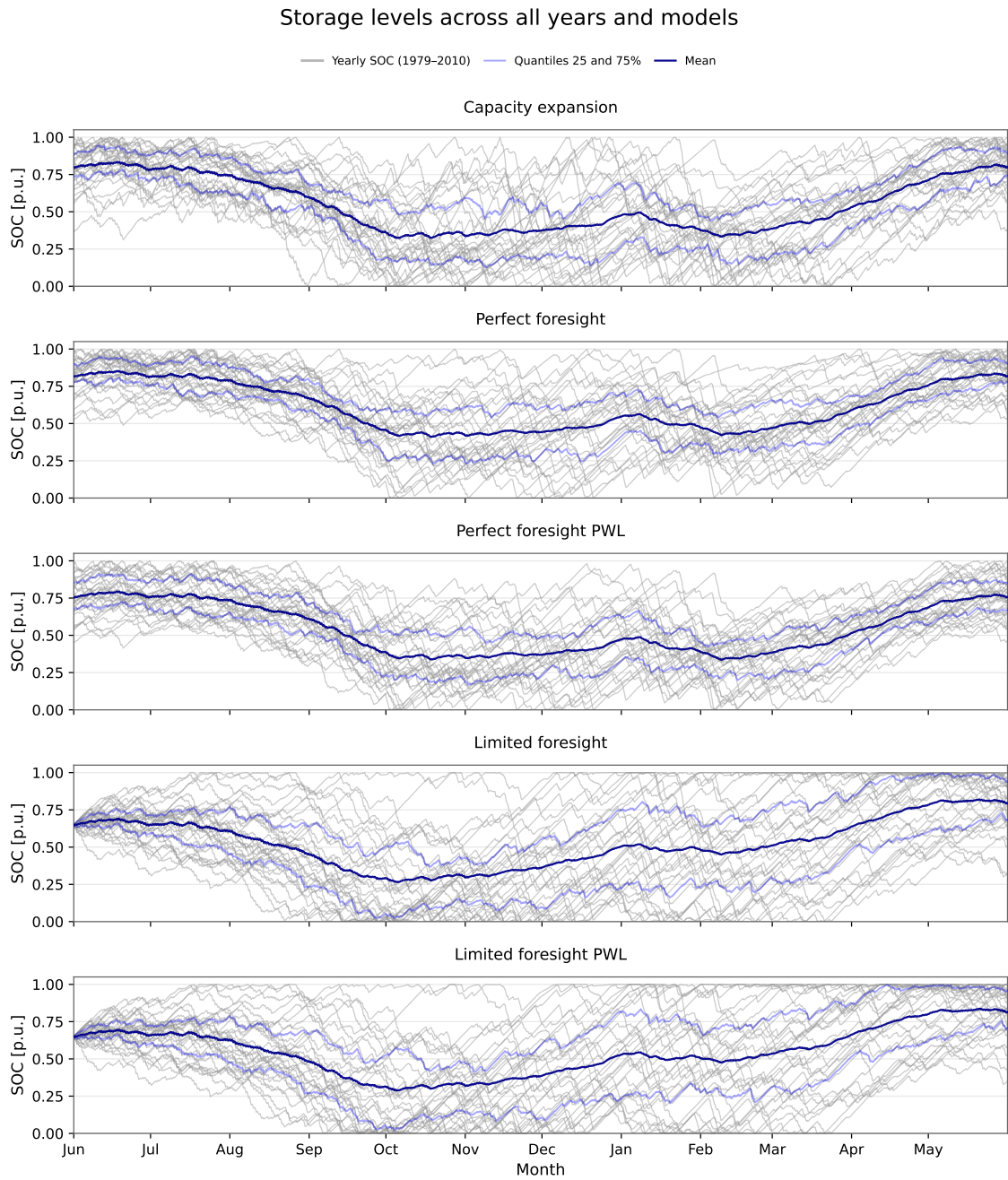


Figure B.7: Yearly SOC operation for all models and years with mean and the 25th + 75th quantiles shown

B.2.2 Yearly storage operation + events for year 2006/2007

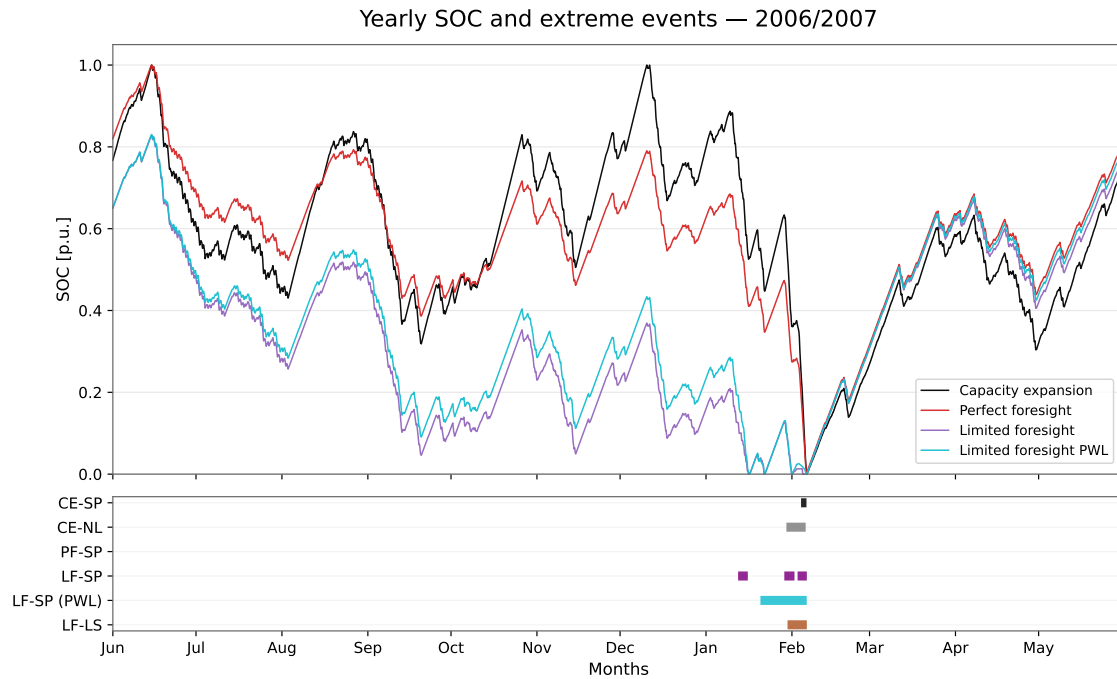


Figure B.8: SOC operations across models with combined with selected events for year 2006/2007.

The abbreviations yield: CE = Capacity expansion, PF = Perfect foresight, LF = Limited foresight, PWL = Piece-wise linear, SP = Shadow price events, NL = Net load event, LS = Load shedding event

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